بسم الله الرحمن الرحيم

وَقُل رَّبِّ زِدْنِي عِلْمًا

EDUCATION WITH HAMZA

SUBJECT:
Chemistry

CLASS:
2nd Year

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تمام تعریفیں ربِ لاشریک کے لئے جس نے مجھ ناچیز کو پیدا کیا۔ اور لاکھوں کروڑوں درود اس ذات پر جس نے ہماری بخشش کے لئے رو رو کر دعا مانگی۔ ان نوٹس کو امت محمدیہ کی آسانی کے لئے بنایا گیا ہے۔ ان کا ثواب ربِ موسیٰ و ہارون نبی آخر زماں ﷺ کی جملہ آل و اولاد اور امت کو عطا کرے آمین۔

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Chp:2 Electrochemistry

Short Questions Answers

1) a) Define Oxidation number. b) How can oxidation and reduction can be defined in term of change in oxidation number?

Ans: a)Oxidation number is the apparent charge positive or negative that an element would have in a compound.

b)Oxidation-Reduction in term of changes in oxidation number:

oxidation and reduction can be defined in term of change in oxidation number:

- An increase in oxidation number is called oxidation.
- A decrease in oxidation number is called reduction.

Example:

$$2Na + Cl2(g) \rightarrow 2NaCl\ (s)$$

* The oxidation number of Na is 0 because it is an elemental form.

$$2Na^{o} + Cl2^{o}(g) \rightarrow 2^{+1}Na\mathit{Cl}^{1-}\ (s)$$

* Oxidation number of Na changes from 0 to +1, so it loses one electron and is called oxidation.

* Each atom in Cl_2 molecule changes its oxidation number from 0 to -1, it gain one electron this is called reduction.

2) Decomposition of H_2O_2 is disproportionation reaction - Justify also Solve the atoms undergoing oxidation or reduction by using oxidation number change method?

Ans: In some redox reactions, the same element atoms is oxidized and reduced at same time. These reactions are called self-oxidation reduction or dis-proportional reactions.

Decomposition of H_2O_2:

$$2{H_2}^{1-}O_2 \rightarrow 2{H_2}^{2-}O + O_2:$$

In this reaction:

* Oxygen is oxidized from -1 in H_2O_2:to 0 in O2.

* Oxygen is reduced from -1 in H_2O_2:to -2 in H_2O.

* Since, O both is reduced and oxidized so it is disproportionation reaction.

b) $6Na^{1+}O^{-2}H^{2+} + 3Cl^{o}2 \rightarrow 5Na^{1+}Cl^{1-} + Na^{1+}Cl^{5+}O3^{-2} + 3H2O^{2(1+)2-}$

* Chlorine is oxidized from 0 in Cl2 to +5 in NaClO3.

* Chlorine is reduced from 0 in Cl2 to -1 in NaCl.

* Since, both oxidation and reduction is brought by Cl so it is disproportionation reaction.

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3) What is construction of Redox Reaction? What is Ion electron method?

Ans: a: In any oxidation, reduction reaction there are two half-reactions:

* The oxidation half-reaction.

* The reduction half-reaction.

- The full redox reaction is sum of these two half reactions.

b: The half-reactions can be balanced by ion-electron method.

- First, equalize the no of electron lost in oxidation with no of electron gained in reduction.
- Then combine the two half-reactions.
- The rest of equation balanced by inspection method.

4) Explain how to identify oxidizing and reducing agents with an example.

Ans: * An oxidizing agent is a substance that causes another substance to oxidize. It has an element that is reduced so it gains electrons.

$$3\overset{0}{P} + 5\overset{+1\ +5\ 3(-2)}{HNO_3} + \overset{2(+1)\ -2}{2H_2O} \longrightarrow \overset{3(+1)\ +5\ 4(-2)}{3H_3PO_4} + 5\overset{+2\ -2}{NO}$$

The HNO3 is undergoing reduction as oxidation state of N changes from +5 in HNO3 to +2 in NO.So HNO3 is an oxidizing agent.

* A reducing agent is a substance that causes the substance to reduce. It has an element that is oxidized so, it loses electrons.

$$\overset{+4}{MnO_2} + 4\overset{-1}{HCl} \longrightarrow \overset{+2}{MnCl_2} + 2H_2O + \overset{0}{Cl_2}$$

* MnO2 is undergoing reduction as oxidation state of Mn changes from +4 in MnO2 to +2 in MnCl2 so MnO2 is an oxidizing agent but HCl is reducing agent because Cl changes from -1 in HCl to 0 in Cl2.

5) a) What is function of salt bridge in galvanic cell? b) Write half reaction occurring at cathode in Zn-Cu cell and write overall reaction.

Ans: a)

1. In one container, a zinc rod is dipped in zinc sulphate solution. It is connected by wire to copper rod dipped in copper sulphate solution in another container.

2. No current flows through external circuit.

* When two solutions are connected with an U-tube, filled with electrolyte solution, current flows through external circuit.

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* The salt bridge allows movement of ions from one solution to other without mixing two solutions.

* Thus, salt bridge keeps electrical neutrality in each half cell.

b) * In Zn-Cu galvanic cell, Zn has greater tendency to lose electrons than Cu. Zn electrode becomes negatively charged compared to Cu electrode. Electrons move from Zn to Cu through external circuit.

* At Anode (Oxidation half-Reaction):

$$Zn_{(s)} \longrightarrow Zn^{2+}_{(aq)} + 2e^-$$

* At Cathode (Reduction half-Reaction):

$$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)}$$

* Over all Reaction:

$$Zn_{(s)} + Cu^{2+}_{(aq)} \longrightarrow Zn^{2+}_{(aq)} + Cu_{(s)}$$

6) a) Which electrode is used as reference electrode for measuring all other electrode potential? b) Write methods to measure electrode potential?

Ans: a) * A standard hydrogen electrode (SHE) is used as reference electrode.

* It has platinum foil coated with finely divided platinum. It is surrounded by H2 gas at 1 atm pressure. It is in contact with 1 M HCl solution at 298K..

b) * Standard hydrogen electrode is used to measure the standard electrode potentials of other half-cells like copper and zinc.

* It has standard electron potential of 0V by definition.

7) Write construction of electrochemical cell? what is the determination of Cell potential?

Ans: * An electrochemical cell setup is in which copper rod is dipped in 1M copper sulphate solution.

* The half cell is connected with standard hydrogen electrode using salt bridge.

* The deflection of voltmeter shows that current flows from hydrogen electrode to copper electrode.

* Electrons move from hydrogen to copper electrode. Copper act as cathode and hydrogen act as anode.

b) A cell reaction is made of two half-reactions.

* Reduction happens in half-cell that has greater value of reduction potential.

* Oxidation happens in half-cell that has smaller value of reduction potential.

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* The equation of half-cell reaction with smaller potential is reversed. The value of reduction potential is then added with greater value of reduction potential.

* The sum of these two equations give equation of cell reaction.

8) a) How is spontaneity of chemical reaction determined? b) Is following reaction feasible Sn + $Fe^{2+} \rightarrow Sn^{2+}$+ Fe? The Standard reduction potential values are E^{o}Sn = -0.14V, E^{o}Fe = -0.44V.

Ans: a) * It can be known from sign of value of E^{o}cell.

* The E^{o}cell is calculated as

$$E^{o}cell = E^{o}cathode - E^{o}anode.$$

* If E^{o}cell.is positive, the reaction occur spontaneously and is feasible.

* If E^{o}cell.is (-), the reaction is not feasible.

b) Solution:

$$Sn + Fe^{2+} \rightarrow Sn^{2+} + Fe$$

* Oxidation of Sn take place: Sn → Sn^{2+}+ $2e^-$

* Reduction of Fe take place: Fe^{2+}+ $2e^-$ → Fe

* E^{o}cell= E^{o}cathode - E^{o}anode.

= -0.44V - (-0.14V) = -0.30V

* Thus E^{o}cell would be (-) so reaction is not feasible.

* However, reverse reaction spontaneous.

9) a) The standard electrode potential of Fe^{3+}/Fe^{2+} is 0.77V. What is electrode potential of system containing 1.0 mol dm^3 of Fe+3+ and 0.2 mol dm^3 of Fe^{+2} ions?

Solution:

$$E = E^{o} + \frac{0.059}{n} log \frac{[oxidized\ specie]}{[reduced\ specie]}$$

$$E\ Fe^{3+}/Fe^{2+} = E^{o}_{\mathbf{Fe3+/Fe2+}} + \frac{0.59}{n} log \frac{[Fe^{3+}]}{[Fe^{2+}]}$$

$$= o.77 + \frac{0.059}{1} log \frac{1.0}{0.2}$$

$$= 0.77 + 0.059 \times 0.099$$

$$= 0.77 + 0.0412$$

$E_{Fe3+/Fe2+}$ =0.811 v

10) a) What is electrochemical series? b) During electrolysis of dilute NaCl solution which ions are preferentially discharged at anode and cathode and why?

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Ans: a) The oxidation potential of an electrode can obtained by reversing sign of reduction potential of that electrode.

b) * The solution contains a lower conc of Cl^- ions. So OH^- ions are discharged at anode. Cl^- ions are not discharged.

* Na^+ ions are more reactive than H2 so they remain in solution & are not discharged. Instead, water molecules discharged at cathode.

11) a) What factors influences products of electrolysis? b) Predict product formed at both electrode when conc. aqueous NaCl is electrolyzed. Explain?

Ans: a) * Electrolysis uses an electric circuit to derive non-spontaneous redox reaction.

* The state of Electrolyte.

* Position in electrochemical series.

* Concentration of electrolyte.

b) * The solution contains high conc. of Cl^- ions. So Cl^- ions are discharged at anode instead of OH^- ion.

* Na^+ are more reactive than the H2 so they remain in solution & are not discharged. Instead H2 molecules discharged at cathode.

12) a) Why does longer electrolysis produce more product even if current is constant? b) what is the significance of relationship F = NA × e.

Ans: a) * The amount of substance produced at electrode during electrolysis depends on current and time for which constant current is passed.

The quantity of charge that passes through electrolyte.

$Q = I \times t$

Q = Quantity of charge in coulomb

I = current in Ampere

t = time of electrolysis in second

b) F is faraday constant

NA is Avogadro constant

e is charge on (e^-)

$F = NA \times e$.

NA = Avogadro constant (no of entities per mole)} = $6.022141 \times 10^{23} mol^-$

E= Elementary charge (charge on single electron) = $1.60217662 \times 10^{-19}$ C. This relation can be used to calculate the charge required to deposit one mole of substance.

13) The oxidation potential of Zn is +0.76 and its reduction potential is -0.76V. b) Na and k displace hydrogen from acids but Cu and Pt cannot.

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Ans a) Oxidation and reduction are reverse to each other. When zinc metal is oxidized, it releases free energy (ΔG > 0).. When zinc ions are reduced, the same amount of energy must be supplied (ΔG > 0).

According to the law of conservation of energy, energy is neither created nor destroyed. It is only transformed from one form to another.

b) Na and k are higher in activity series than SHE, so they donate electrons and thus displace H^+ from acid readily.

Cu and Pt are low in activity series than SHE, so they do not displace hydrogen from acids.

$$2Na + 2HCl \longrightarrow 2NaCl + H_2$$
$$2K + 2HCl \longrightarrow 2KCl + H_2$$

(14) a) What is oxidation no. of oxygen in H_2O_2? State purpose of Winkler method.

Ans a) Let x = Oxidation number of O.

Then for H_2O_2:

2(+1) + 2x = 0

2x = -2

x = -1

In H_2O_2, O has oxidation number of -1 because it is part of peroxide.

b) It is used to determine dissolved O2 content in H2O sample. Oxygen in water sample oxidizes iodide ions into Iodine. The volume of thiosulphate used in titration gives the amount of original O2 in water sample.

(15) a) What is significance of activity series of metals? b) What is oxidation number of chromium in Cr^2 O_7^{2-}?

Ans a) It shows reactivity order of metals. It can predict which metal can displace another from its compound. Mg is above Zn in activity series, so it has more tendency to give electrons; therefore, it displaces Zn^{2+}

Mg + ZnSO4 → MgSO4 + Zn

b) Let x = oxidation number of Cr.

2x + 7(-2) = -2

2x - 14 = -2

2x = 12

x = +6

(16) Differentiate b/w Galvanic cell and Electrolytic cell?

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Galvanic Cell	Electrolytic cell
• It converts chemical energy into electrical energy. • The anode is (-) and cathode is (+). • Example: Zn-Cu Daniel cell producing electricity.	• It converts electrical energy into chemical energy. • The anode is (+) and cathode is (-). • Example: Electrolysis of fused NaCl to obtain Na & cl2

(17) Differentiate b/w oxidation half reaction and reduction half reaction?

Oxidation H.R	Reduction H.R
• It involves loss of electron • Electrons appear on product side. • Oxidation no. increases.	• It involves gain of electrons. • Electrons appear on reactants side. • Oxidation no. decreases

(18) a) Explain why anode is negative in galvanic cell but positive in electrolytic cell? b) Why MnO2 is called oxidizing agent in reaction: MnO2 + 4HCl → MnCl2 + Cl2 + 2H2O?

Ans a) In a galvanic cell, chemical oxidation is spontaneous. Oxidation occurs at the anode where electrons are released and flow through the external circuit towards the cathode.

In an electrolytic cell, the reaction is non-spontaneous. It is driven by an external power source (battery) which pulls electrons away from the anode, causing oxidation to occur.

b) MnO2 helps chlorine form from Cl^-. Cl^- is oxidized to Cl2. MnO2 is reduced to Mn^{2+}. A substance which is reduced is an oxidizing agent.

Chp: 3 Chemical Equilibrium

(1) What is base dissociation constant? How is it related to base strength?

Ans: Strength of base is the ability to accept a proton from solvent e.g., H2O.

For a base B, an equilibrium reaction with H2O can be represented as:

$B + H_2O \rightleftharpoons BH^+ + OH^-$

$Kc = \frac{[BH^+][OH^-]}{[B][H2O]}$

Or $Kc[H_2O] = Kb \frac{[BH^+][OH^-]}{[B]}$

Kb is called Base Ionization Constant.

(2) Explain significance of pKb values?

Ans: Kb values are generally inconvenient. Thus for convenience, Kb values are converted to pKb values. pKb is the negative logarithm of Kb.

pKb = -log Kb

Higher Kb means lower pKb and a stronger base.

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(3) a) How can an acidic and basic buffer be prepared? Give an example? b) How does a buffer work upon addition of acid or base?

Ans a): By mixing a weak acid with a salt of it with a strong base. Such solution gives acidic buffers with pH less than 7, e.g., Acetic acid.

By mixing a weak base and a salt of it with a strong acid. Such solution gives a basic buffer with pH more than 7, e.g., NH4OH with NH4Cl.

Consider a buffer solution of CH3COOH and CH3COONa. The working of a buffer can be explained by the common ion effect. CH3COOH is a weak electrolyte and undergoes very little dissociation. CH3COONa is a strong electrolyte. When CH3COONa is added to CH3COOH solution, the dissociation of CH3COOH is decreased due to the common ion effect of $CH3COO^-$.

(4) a) What happens when an acid is added to acetic acid - sodium acetate buffer? b) What happens when a base is added acetic acid - sodium acetate buffer?

Ans a) **Addition of Acid:** Suppose a few drops of HCl are added to it. Its H^+ ions are used up by CH3COO- ions. Thus, upon addition of HCl, the pH of the buffer solution will not change.

$$CH3COO^- + H^+ \rightarrow CH3COOH$$

b**) Addition of Base:** In the same buffer solution, if a strong base is added, it is neutralized by the acid.

$CH3COOH \rightarrow CH3COO^- + H2O$

(5) How does carbonate/bicarbonate buffer help maintain blood pH?

Ans: The bicarbonate ions play a role in maintaining blood pH. Blood pH measures acidity or basicity and must be tightly regulated. The normal pH range of blood is approximately 7.35–7.45. A value lower than 6.8 or higher than 7.8 can lead to death. In the bicarbonate buffer system, CO2 and $HCO3^-$ exchange reversibly.

a) Formation of Carbonic Acid: $CO2 + H2O \rightleftharpoons H2CO3$

b) Decomposition of Carbonic Acid: $H2CO3 \rightleftharpoons H^+ + HCO3^-$

(6) Why are buffers important in soap and food industry?

Ans: Buffers control acidity in blood and maintain stable pH in pool water. In industry, buffers prevent skin irritation and make products gentle.

(7) a) Calculate concentration of AgCl in aqueous solution if Ksp = 1.8×10^{-10} at 25° C? b) Ksp for BaSO4 is $1 \times 10^{-10.}$ Write equation of equilibrium expression for dissolving of barium sulphate?

Ans a) The dissociation equilibrium is: $AgCl(s) \rightleftharpoons Ag^+(aq) + Cl^-(aq)$

Ksp expression: $Ksp = [Ag^+][Cl^-]$

Let solubility of AgCl be x mol/dm^3. Each mole of AgCl gives one mole of Ag^+ and Cl^- ions.

$AgBr(s) \rightleftharpoons Ag^+ + Cl^-$

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$Ksp = [X][X]$

1.8×10^{-10}

$x = 1.34 \times 10^{-5}$ mol/dm^3

b) The dissociation equilibrium is:

$BaSO4 \rightleftharpoons Ba^{2+}(aq) + SO4^{2-}(aq)$

Ksp expression: $Ksp = [Ba^{2+}][SO4^{2-}]$

(8) a) State Nernst Distribution Law. What is Partition Coefficient with explanation? b) What are calculations of Partition Coefficient?

Ans a) When a solute is distributed between two immiscible liquids at constant temperature, the ratio of its concentration in the two liquid layers is constant. It is called the Partition Coefficient (Kp).

It is the ratio of concentrations of solute in two immiscible solvents when equilibrium is established at a particular temperature.

Explanation: Two immiscible liquids, like ether and H2O, are put in a separatory funnel and shaken two layers form upon standing. Ether is less dense than H2O, so it forms the upper layer.

- Let a solute X dissolves in both.
- Measure the concentration of X in water layer it will decrease..
- The partition coefficient Kp for the system where the solute is in the same physical state in both solvents can be calculated using the equilibrium expression.

(9) a) How does polarity of solute influence its distribution b/w polar and non-polar solvents? b) How does temperature affect partitioning of solute b/w two liquids?

- Ans a) Polarity of Solute
- Polarity of Solvent.
- A polar solute prefers the polar phase.e.g. water.

* They interact with polar solvents through hydrogen bonding.

Example: Ethanol has a polar hydroxyl group.

* Polar solvents favour polar solutes, while non-polar solvents favour non-polar solutes.

Example: Benzene is non-polar.

b) Temperature changes the solubility of solutes in both solvents and thus K.E. Generally, at higher temperature, solubility increases in polar solvents.

Example: Aspirin in a polar solvent.

(10) Explain Common ion effect with example? b) Differentiate b/w strong and weak acid using extent of Ionization and Ka?

Ans a) The Degree of ionization or solubility of an electrolyte is suppressed by adding a highly soluble electrolyte containing a common ion.

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In solution of weak acid HF, adding NaF increases the F^- ion conc. According to the Le Châtelier's Principle, the equilibrium shifts left:

$$HF \rightleftharpoons H^+ + F^-$$

The dissociation of HF decreases and increases its concentration.

b) * Strong acids almost ionize completely in H2O. It gives more H^+ ions and have a very large Ka.

* Weak acids ionize partially, giving fewer H^+ ions and have small Ka.

The larger the Ka value, the stronger the acid.

Example: HCl is a stronger acid than acetic acid. So HCl has a larger Ka than acetic acid.

(11) a) What does a Kpc greater than 1 indicate? b) Differentiate b/w acidic buffer & basic buffer?

Ans: a) If Kpc > 1, it means concentration of solute is higher in organic layer than aqueous layer. Thus the solute is more soluble in organic solvent. This concept is important in drug design and extraction process. The drug with Kpc > 1 means that it has more solubility in actanol.

Acidic Buffer	Basic Buffer
The pH of an acid buffer is less than 7. e.g., CH3COOH. It resists change in pH when base is added.	The pH of basic buffer is greater than 7. e.g., NH4Cl. It resists change in pH when acid is added.

(12) a) Why is partition coefficient important in drug design? b) Explain in detail how common ion effect decrease solubility of slightly soluble salts and relate it with Le Chatelier's principle?

Ans: a) The partition coefficient shows how a drug distributes b/w aqueous (blood) and lipid (membrane) phases. Drugs with very low lipid solubility cannot cross cell membrane effectively.

b) Common ion effect is decrease in ionization of weak electrolyte by adding a strong electrolyte having a common ion. It also decreases solubility of slightly soluble salts.

Example: The solubility of AgCl decreases in presence of NaCl.

$$\mathbf{AgCl \rightleftharpoons Ag^+ + Cl^-}$$

Addition of NaCl increases Cl- ions:

$$\mathbf{NaCl \rightleftharpoons Na^+ + Cl^-}$$

According to Le Châtelier's Principle, the equilibrium shifts to the left, decreasing solubility.

(13) a) Differentiate b/w Solubility and Solubility product? b) Differentiate b/w strong base and weak base?

Ans:

Solubility	Solubility Product

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1- It is product of molar conc of ions in saturated solution of soluble salt.	1- It is the no of moles of solute which dissolves in one dm^3 of solution.
2- Its value is constant at constant temperature for a given salt.	2- Its value changes when temperature changes.
3- It is represented by: Ksp = $[cation]^a [anion]^b$.	

Strong Base	Weak Base
1- A strong base completely ionizes in aqueous solution.	1- A weak base partially ionizes in aqueous solution.
2- It has very large value of Kb and small value of pKb.	2- It has very small value of Kb and large value of pKb.
3- Its pH is very high for same molar concentration	3- Its pH is lower for same molar concentration.

(14) a) Why is HCl not as strong an acid in acetic acid as in water? b) Which has larger value of pKa: a strong acid or weak acid?

Ans a):* In H2O, HCl is completely ionized because water is a good base for HCl. It accepts protons from HCl readily. Thus, in water4, HCL behave as strong acid.

* In acetic acid, HCl is not completely ionized. Acetic acid is less basic than water. Due to weak basic nature, it will not readily accept proton from HCl. Thus, HCl in acetic acid behaves as a weak acid.

b) pKa is defined as negative logarithm of Ka (pKa = -log Ka). pKa and Ka are inversely proportional to each other.

* Strong acids have greater Ka and smaller pKa values.

* Weak acids have smaller Ka and greater pKa values.

Chp 4: Acid-Base Chemistry

(1) What is the purpose of titration and how it is performed? b) Differentiate b/w Equivalence and end point?

Ans a):

- To find unknown concentration of solution, titration is used.
- It is done in neutralization reactions between acids and alkalis.
- A titrant of known concentration is added from a burette to a conical flask containing the analyte of unknown concentration.
- An indicator is added; the indicator changes colour at the end point of titration.
- The end point is when equal to moles of titration and analyte reaction.
- The equivalence point is at middle of vertical part of curve.

End point	Equivalence point

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The endpoint of titration is reached when the reaction is complete due to formation of product and indicator changes color.	It is the point in titration at which amount of titrant added is just enough to neutralize the analyte solution completely.
It is point where colour change occur in system.	It is point where chemical reaction comes to end.
It is a less precise measurement.	It is more precise measurement.
It is macroscopic event	It is microscopic event

(2) What happens to pH titration when strong acid is titrated against strong base?

Ans: * HCl is taken in conical flask. At first, only H^+ ions are in flask; initial pH is about 1.2.

* Strong alkali (NaOH) is added from burette, OH^- ions react with H^+ ions to form water. The pH increases slightly at first.

* When the volume of NaOH approaches equivalence point, pH rises steeply.

*** At equivalence point, pH = 7.**

(3) a) Why is the equivalence point acidic in a strong acid-weak alkali titration? b) Why is it basic in a weak acid-strong alkali titration?

Ans a):

* Initially, only H^+ ions are in the flask (pH is about 1.2).

* Weak alkali (NH3) is added, H^+ ions react with NH3. The pH rises slowly at first.

* At equivalence point, the solution is not neutral but still acidic (pH is about 5.5).

* This happens because H^+ reacts with NH3 to form $NH4^+$; $NH4^+$ is a relatively strong acid, so the solution stays acidic.

b): * Ethanioc acid (CH_3COOH) is taken in conical flask; initial pH is about 2-3.

* NaOH is added; H^+ ions react with OH^- ions from NaOH to form water.Near equivalence point, pH rises very steeply.

* The equivalence point is not neutral; the solution is slightly basic (pH is about 9).

(4) a) Why is there no vertical region in a weak acid-base titration curves? b) Why does the final pH not reach 14 in titration of weak base with strong acid?

Ans a): In weak acid-weak base titration, both reactants are partially ionized. The change is gradual and pH changes throughout the titration. There are no sudden or sharp changes in pH near the equivalence point, so no steep vertical region is observed. However, curve shows a smooth inflexion point.

b) When a weak base (NH3) is titrated with a strong acid, excess NH3 dominates before equivalence. However, since NH_3 is not a strong base, the initial pH values are lower than those observed with strong bases. Therefore, the final pH does not reach 14 but stays below it.

(5) a) Why is there a buffer region in titration curve of weak acid with strong base? b) Explain shape of pH titration curve for strong acid against strong alkali?

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Ans a): Before the equivalence point, both the weak acid and its conjugate base are present. They resist changes in pH when small amounts of base are added, forming a buffer solution. This causes a gradual instead of a sudden jump in pH.

b) Initially, only H^+ ions are present; pH is very low (1-2). As strong base (NaOH) is added, H^+ and OH^- react: HCl + NaOH → NaCl + H2O. pH rises slowly at first, then sharply near the equivalence point (pH = 7). Adding more NaOH gives a pH of 13-14. The curve shows a steep vertical section in the middle.

Chp: 5 Group 2 Elements

(1) a) What are alkaline earth metals? Why are these called so? b) What is electronic configuration of first three Group 2 elements?

Ans a): The elements in Group 2 of the periodic table are called Alkaline earth metals. They include Be, Mg, Ca, Sr, Ba, Ra.

* They are called so because their oxides and hydroxides are alkaline (water soluble) and Their oxides are found in the earth's crust.

* These elements are in Group 2, so they have two electrons in their outermost shell.

these two outer electrons occupy S subshell.

b) the general valence shell electronic configuration in ns^2 .Electronic configuration of first three groups 2 elements are:

beryllium (4Be)= $1s^2$, $2s^2$

magnesium (12Mg)= $1s^2$, $2s^2$, $2p^6$, $3s^2$

calcium (20Ca) = $1s^2$, $2s^2$, $2p^6$, $3s^2$, $3p^6$, $4s^2$

(02)a) How does the reactivity of group 2 metals change down the group and why? (b) Explain the trend of atomic radius down the group?

Ans) a: The first and 2nd ionization energies of group 2 elements decrease down the group, and it become easier to remove two outer electrons of metals going down the group. It is due to increased shielding effect and larger distance b/w outermost electrons and nucleus. so it gets easier for atoms to lose two electrons and form 2+ ions. Thus, elements become more reactive down the group.

(b) The Atomic radius increases down the group. It is because down the group, no of shells and shielding effect increases. Thus attraction b/w nucleus and valence electrons decreases. Hence, atomic radius increases.

* Hence, increasing order of Atomic radius of group 2 element is Be < Mg < Ca < Sr < Ba

(3) Prove that BeO is Amphoteric? b) What is flame test? what information can it give?

Ans: a) All oxides of Group 2 are basic except BeO.

* BeO is Amphoteric because it can react with both acid and base.

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* With acid:

$BeO(s) + 2HCl(aq) \rightarrow BeCl_2(aq) + H_2O(l)$

* With base:

$BeO(s) + 2NaOH(aq) \rightarrow Na_2BeO_2(aq) + H_2O(l)$

b) Some Group 2 elements burn with characteristic flame colours. The 2+ ions formed in the flame cause these colors. Flame tests can identify calcium, strontium and Barium compounds.

Procedure:

* A nichrome wire is cleaned with Conc. HCl.

* The wire is dipped into salt sample.

* It is then heated in non-luminous Bunsen flame.

Following flame colours may be observed:

* Calcium compounds: brick-red colour

* Strontium compounds: scarlet-red color

* Barium compounds: apple-green colour

(04) a) How magnesium sulfate and calcium chloride be prepared? What are their uses? b) What is general trend in melting points going down Group 2? (c) Explain why any atoms in Group 2 as in other group, get larger with increasing atomic number?

Ans: a) (Group 2 oxides are basic oxides. They are used to make useful compounds by reacting with acids.

* $MgO(s) + H_2SO_4(aq) \rightarrow MgSO_4(aq) + H_2O(l)$

* $CaO(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l)$

Uses of Magnesium sulfate:

* Used for short term relief of constipation.

* Used for a soaking solution for sprains, bruises, muscle aches, discomfort and joint stiffness.

Uses of Calcium chloride:

* Used as solidify agent in paint production

* Used as coagulant in manufacture of rubber.

(b) The melting point of group 2 elements decreases down the group. Atomic size increases, so attraction b/w nucleus and delocalised electrons decreases. Therefore, metallic bond becomes weaker and melting point decreases.

C) The atomic radius increases down the group. So No of shells and shielding effect increases. attraction b/w nucleus and valence electron decrease, and atomic radius become larger.

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(05) a) Compare the reactivity of group 1 and group 2 elements with water. b) what are uses of lime stone in industry?

Ans: a) Alkali metal react vigorously with water. The reaction is so violent that the evolved hydrogen gas catches fire.

* $2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$

* Alkaline earth metals react slowly with water.

* $Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g)$

(b) Lime stone is mainly made up of calcium carbonate ($CaCO_3$). There are many types of limestone which provide useful rocks.

* Marble is another form of calcium carbonate. It is used for building purposes. e.g making expensive tiles.

* Most calcium carbonate is used to make cement.

(06) a) How does slaked lime helpful in neutralizing acidity of soil? b) What are uses of group 2 compounds in medicines?

Ans: **a) Uses of slaked lime:**

Calcium hydroxide (slaked lime) is used in agriculture. It is used to increase the pH of acidic soils. Calcium hydroxide is basic. It react with and neutralizes acidity of soil and pH of soil is increased.

b) Uses of Group 2 Compound in medicine.

Barium sulphate ($BaSO_4$) helps to diagnose stomach and intestine disorders. It absorbs X-rays strongly. It is insoluble, so it is not absorbed into blood stream.

* Gaviscon contains calcium carbonate and magnesium carbonate.

* Rennies contains calcium carbonate and magnesium carbonate.

* Milk of Magnesia suspension of magnesium hydroxide in H_2O.

Reaction of Antacids with stomach acid:

* $CaCO_3(s) + 2HCl(aq) \rightarrow CaCl_2(aq) + H_2O(l) + CO_2(g)$

* $Mg(OH)_2(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + 2H_2O(l)$

(07) a) Why thermal stability of Group 2 increases down group? b) Write a balanced chemical equation, including the state symbols, for the thermal decomposition of i- strontium carbonate ii- Barium nitrate.

Ans: a) The carbonate ion (CO_3^{2-}) has large ionic radius. Large ion is easily polarised by a small and highly charged cation. Down the group 2, the size of cation increases: $Mg^{2+} < Ca^{2+} < Sr^{2+} < Ba^{2+}$. Therefore, polarizing power decreases and thermal stability increases down the group.

(b) i- Strontium Carbonate: $SrCO_3(s) \rightarrow \Delta SrO(s) + CO_2(g)$

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ii- Barium nitrate: $2Ba(NO_3)_2(s) \xrightarrow{\Delta} 2BaO(s) + 4NO_2(g) + O_2(g)$

(08) What is the trend in solubility of hydroxides of group 2 elements?

Ans: The solubility of hydroxides increases down the group. Barium hydroxide is highly soluble in water.

* The order of solubility is $Mg(OH)_2 < Ca(OH)_2 < Sr(OH)_2 < Ba(OH)_2$.

* The variation in solubility can be explained by comparing values of enthalpy change of hydration and lattice energy.

(09) What do you mean by roasting and reduction in metallurgy process?

Ans: **1- Extraction: a) Ore Prepration**

* Group 2 elements are usually found in minerals like carbonates, sulfates, oxides and silicates.

* The Ore is mined, crushed and grounded to a fine powder in order to extract elemental metal.

b- Roasting

* Group 2 Carbonates are unstable on heating. They decompose to give a metal oxide and CO_2.

* Example: to extract magnesium from its carbonate, it is first roasted: $MgCO_3(s) \rightarrow MgO(s) + CO_2(g)$

c- Reduction

* On ore, reduction is carried out to extract the metal.

* Depending on ore, reduction can be done by electrolysis of molten oxide.

* Example: Magnesium can be extracted from magnesium oxide.

(10) a) Why is beryllium different from other Group 2 elements in terms of its chemical behaviour. b) Explain why Group 2 metals act as good reducing agents?

Ans. a) Beryllium has very small atomic radius and very high polarizing power compared to other Group 2 elements. Thus, it has more covalent characters in its compounds. e.g: BeO and $BeCl_2$.

* BeO is amphoteric while other oxides are basic.

* Be forms complexes such as $[Be(OH)_4]^{2-}$, while other member do not form.

b) Group 2 metals have two electrons in their outermost subshell. They can lose these two electrons to easily form M^{2+} ions because their ionization Energies are relatively low. The ability to lose electrons means they themselves get oxidized and help reduce other species. Their reducing strength increases down the group due to decreasing ionization energy.

(11) Why do Group 2 metals form M^{2+} ions but M^{3+} ions not?

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Ans: Group 2 metals have two electrons in outermost shell. It is easy to remove both because they are far from the nucleus and feel weak attraction. After loosing two electrons, the metal gets stable noble gas configuration. That is why M^{2+} ions are more stable than M^{3+} ions.

(12) Why BeCl_2 is covalent while BaCl_2 is ionic?

Ans: Be^{2+} is very small ion. It has high polarizing power and attracts electron cloud of Cl^- strongly. So much so covalent bond forms. Ba^{2+} is much larger and cannot polarize Cl^- so BaCl2 is ionic.

Chp: 6 The Transition Metals

(1) a) Define transition elements and Explain why they are called transition? (b) Why is zinc not considered a transition element even in the d-block?

Ans: a) Transition elements are those which show an incomplete d-subshell through most stable ionic states.

* They are called transition because they show a gradual change from very reactive group 2 Elements (Calcium) to much less reactive group 13 elements (gallium).

* The first series of transition element starts from scandium to zinc.

(b) Zinc does not follow definition of transition elements.

Zinc is a d-block element. So it has a complete 3d subshell. It forms only one colourless ion, Zn^{2+}, with valence electrons in d orbital.

(2) a) Explain why chromium and copper have exceptional electronic Configuration. b) What are the relative energies of 4s and 3d subshells?

Ans a) Transition metals follow Aufbau principle. But chromium and copper are exceptions. The electronic Configuration of Cu and Cr are:

$_{24}Cr$ [Ar] $3d^5\ 4s^1$

$_{29}Cu$ [Ar] $3d^{10}\ 4s^1$

It is because:

* Atoms or ions with half-filled 3d ($3d^5$) or completely filled 3d ($3d^{10}$) configuration are more stable.

* To get stable configuration, one electron is less in 4s orbital as compare to other transition elements.

(b) The energy difference b/w 3d and 4s Orbitals changes across transition metals.

*In Calcium, 4s has lower energy than 3d. Because 4s electrons feel strong attraction of unscreened nucleus.

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* Across period after calcium, 4s electrons penetrate more than 3d. 3d are shielded by 3d electrons. This decreases their energy slightly. It also cause a small rise in first and second ionization energies.

(3) a) Write six physical properties of transition elements? b) why density generally increases from left to right in transition elements?

Ans a)i) They have high melting points because of strong metallic bonding in them.

ii) They have high density due to close packing of atoms in their lattices.

iii) They are good conductors of electricity because they have free electrons.

iv) They are good thermal conductors.

v) They have shiny metallic luster.

vi) They are malleable and ductile.

(b) * Density generally increases across first period of transition metals from Scandium to copper. This is due to factor:

i- The atomic mass increases from left to right so nuclei become heavier.

ii- The atomic radius decreases due to increasing nuclear charge. This pulls electron cloud closer, reducing the size of atoms.

(4) a) How does atomic radii vary from left to right in first row transition elements and why M.P and B.P of transition element are higher than s-block and p-block?

Ans a) Atomic radii decreases across period because nuclear charge increases and pull electrons closer to nucleus. and increasing nuclear charge is stronger than shielding effect. In middle, the decrease in radii is not very strong because nuclear charge and shielding effect almost remains constant.

b) Because their metallic bonding involves both 3d and 4s electrons. Metallic bonding depends upon no of unpaired es in outer shells.

- melting and B.P increases as atomic number increases.

(5) a) Why transition elements show paramagnetism? b) ferromagnetic substance are much strongly attracted by magnet than paramagnetic substance why and difference between paramagnetic and ferromagnetic?

Ans a) They are weakly attracted by a magnetic field.e.g. Na, Al.

- They have unpaired electrons in valence shells. These electrons are not influenced by surrounding atoms. They point randomly.
- When they are placed in magnetic field, the magnetic moments align in same direction. This cause weak attraction.

(b) They are strongly attracted in magnetic field. e.g Ni, Fe.

* They have unpaired electrons in valence shells. Their unpaired es are influenced by neighboring atoms and align in same direction.

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* When placed in magnetic field, they show very strong attraction, almost millions times stronger than paramagnetic substances.

Paramagnetic Substances	Diamagnetic Substances
They are weakly attracted in magnetic field.	They are strongly repelled by magnetic field.
They have unpaired electrons in their valence shells.	They have paired electrons.
Example Al, Na	Example Fe, Co, Ni

(06) a) How does variable oxidation state help transition element as a catalyst? b) How does vacant d-orbitals help transition elements to act as catalyst?

Ans **a) Variable Oxidation state**

Transition elements can exist in multiple stable oxidation states. First row transition metals can easily change b/w oxidation states. e.g. Fe^{2+}, Fe^{3+} and Cu^{+}, Cu^{2+}. During reaction they gain or lose electrons temporarily. This help reactants convert into products.

b) Vacant d orbital

Transition elements have incomplete 3d Orbitals. These orbitals can accept electron pair from ligands. Ligands form temporary dative bonds with transition metals. These bonds hold reactants on surface of metal, bringing them closer. This weakens the reactants bond and help them break easily.

(7) a) Differentiate b/w Homogeneous and Heterogeneous catalysis?

Homogeneous Catalysis	Heterogeneous Catalysis
These are examples of homogeneous catalysis because reactants and products are same states	These are examples of heterogeneous catalysis because reactants and catalyst in different states.
Oxidation of iodide by peroxydisulphate ions. $2I^{-}(aq) + S_2O_8^{2-}(aq) \rightarrow I_2 + 2\,SO_4^{2-}$	In Haber process, ammonia synthesis iron catalyst absorbs N_2 and H_2 on its surface and makes them react faster.

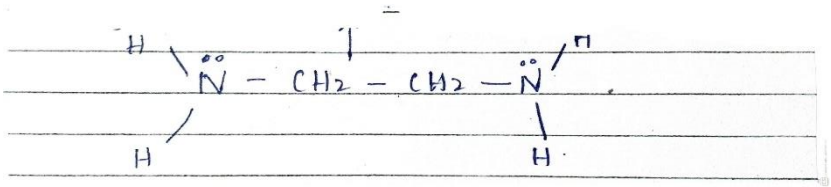
(8) a) What are monodentate ligands with examples b) what are polydentate ligands with examples?

Ans **a) Monodentate ligands:**

They can form one dative bond with metal ion. They have more than one lone pair, but only one pair is donated. e.g: H_2O (Aqua), OH^{-} (Hydroxo), Cl^{-} (Halo).

b) Polydentate ligands:

They can donate more than one lone pair. They form two dative bonds with central ion at same time. e.g: 1,2-diaminoethane ($NH_2CH_2CH_2NH_2$).



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(09) a) How does copper ions form complex with chloride ligands? b) How does cobalt ions form complex with water ligands?

Ans a) With Chloride Ligands

Copper ion mixes 4s and 4p orbitals, giving four sp^3-hybrid orbitals. Four Chloride ions form dative bonds with these orbitals of copper ion. The complex is tetrachlorocuprate ion. The shape is tetrahedral. It is because large size chloride ions cannot fit around a small copper ion.

b) with water Ligand:

Cobalt has configuration $[Ar]3d^7\ 4s^2$. Cobalt ion has configuration $[Ar]3d^7$. Six water molecules coordinate with cobalt ion. Six sp^3d^2 hybrid orbitals form, and six coordinate bonds are formed. The complex is octahedral called hexaaquacobalt ion.

(10) a) Define coordination number with example b) What is coordination sphere Give example?

Ans a) coordination number:

The number of coordinate covalent bonds formed by metal ions with ligands is called coordination number. The most common coordination numbers are 2, 4 and 6.

Example: $[Cu(NH_3)_4]^{2+}$ has coordination number 4. $[Ni(CO)_4]$ has coordination number 4.

b) Coordinating Sphere

The Central atom with Ligands is called coordinative or coordination sphere enclosed by square bracket. The coordination sphere may be neutral, cationic or anionic.

Example: $K^+\ [Fe(CN)_6]^-$ has an anionic coordination sphere.

$[Ni(CO)_4]$ has a neutral coordination sphere.

(11) a) Explain why copper ion can bond with four chloride ligands, but with six water and ammonia ligands? b) Describe geometry and bond angles of the complexes with coordination number 6.

Ans a) Copper ion can bond to four chloride ligands, because chlorine is bigger ligand than oxygen and nitrogen. Chloride belongs to period 3 while water and ammonia belongs to period 2.

b) These complexes adopt octahedral shapes. Octahedral complexes are common and stable.

* Almost all transition metals can form Octahedral complexes.

* Both monodentate and polydentate ligands can give.

* Six donor atoms occupy vertices of an Octahedron.

* Angles b/w octahedral complexes are 90°.

(12) a) What are Ligands Substitution reaction? Give an example. b) What happens when excess NH_3 is treated with $[Cu(H_2O)_4(OH)_2]$ solution?

Ans a) The reaction in which one ligand is substituted by another ligand is called ligand substitution reaction, also called ligand exchange reaction. e.g:- H_2O, NH_3, OH^-, Cl^-.

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b) Adding excess NH_3: dissolve the pale-blue precipitate to give deep blue solution.

$[Cu(H_2O)_4(OH)_2] + 4\ NH_3 \rightarrow [Cu(NH_3)_4(H_2O)_2]^{2+} + 2H_2O + 2OH^-$

(pale blue precipitate) → (deep blue solution)

(13) a) Why do transition elements show variable oxidation state? b) Why Ag^+ ion is more stable than Ag^{2+} ion?

Ans a) Because they have d-electrons in addition to s electrons for bond formation. These elements have ns and (n-1)d electrons. The energies of ns orbitals and (n-1)d are very closer to each other. Thus, (n-1)d electrons are as easily lost as an ns electron. Iron show several oxidation states common are Fe (II) and Fe (III).

b) The electronic Configuration of $_{47}Ag$ is [Kr] $4d^{10}\ 5s^1$. Thus the configuration of Ag^+ and Ag^{2+} will be:

Ag^+ [Kr] $4d^{10}\ 5s^0$

Ag^{2+} [Kr] $4d^9\ 5s^0$

In Ag^+, 4d sub-shell is completed. In Ag^{2+}, 4d subshell is not complete. It is partially filled. A completely filled subshell is always more stable. Hence Ag^+ is more stable than Ag^{2+}.

Chp:1 Organic Chemistry

(1) a) Define Isomerism? b) Differentiate between Constitutional isomers and sterioisomers?

Ans: a) The compound with same molecular formula but different structural formula and property.

Constitutional Isomers	Stereoisomers
1) They have different connectivity of atoms in molecule	1) They have same connectivity but different three dimensional arrangement of atoms in space.
2) These can only be interconverted by breaking and reforming of chemical bonds	2) They may be interconverted by rotation around single bonds or may be interconverted by breaking and reforming of chemical bonds
3) Example: n-butane and isobutane.	3) Cis-2 butene and trans-2 butene.

(2) a) Define Stereoisomerism b) Differentiate b/w conformational and configuration Isomerism.

Ans: a) Molecules having same molecular formula and same structural formula but different arrangement of atoms in space.

Conformational isomerism	Configuration isomerism
In this isomerism, Stereoisomers are Interconverted simply by rotation around single bond.	In this isomerism Stereoisomers are interconverted by breaking and remaking of chemical bonds.

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(3) a) Define Geometrical isomerism? why alkenes show geometrical isomerism b) Alkanes cannot show geometrical isomerism? why.

Ans: a) They are differ in geometric arrangement of atoms or groups around double bond. Such molecules are called geometric isomers and phenomenon is called Geometric isomerism.

In alkenes, each carbon atom of double bond is sp^2 hybridized. The double bond contains one sigma formed by (head on head overlap of sp^2 orbital. One pi bond formed by (sideways of unhybridized pz orbitals). The pi-bond in alkenes restricts rotation around double bond.

b) Alkanes cannot show geometric isomerism because single bonds can rotate freely. Rotation around C-C single bond in ethane requires only about 12 kJ/mol energy which is available at room temperature.

(04) a) Why cis- isomer is more polar than trans isomer? b) Why cis- isomer more reactive than trans- isomer?

Ans: a) Cis and trans isomers have different physical properties such as melting points, B.P and solubility. Trans-isomers are generally non-polar with zero dipole moment. It is because individual bond moments cancel the effect of each other which are in opposite direction.

b) Cis-trans isomers show differences due to different shapes. Cis isomers are usually more reactive than trans- isomers.

(05) a) Define optical Isomerism? b) Differentiate b/w Chiral and Achiral molecules?

Ans: a) An optically active compound can exist in two forms which rotate plane polarized light in opposite direction. They are known as Optical isomers and phenomenon is optical isomerism.

Chiral molecules	Achiral molecules
Organic molecules may exist as non-superimposable mirror images. These have carbon attached to four different groups eg. bromochlorofluoro methane..	Achiral molecules have carbon attached to identical groups. Achiral mirror images are superimposable.

(06) a) Differentiate b/w Enantiomers and diastereomers? b) Define racemic mixture?

Ans:

Enantiomers	Diastereomers
They are non-superimposable mirror images of each other. Perfect mirror images.	Not mirror images. They are not mirror images of each other.
They occur in molecules which contains at least one chiral center	They have different properties e.g: melting, boiling point, solubility.
(+)-tartaric acid and (-)-tartaric acid.	(+)-tartaric acid and meso-tartaric acid.

b) A mixture containing 50% of each enantiomer is called racemic mixture. Racemic mixture has equal dextrorotatory(+) and levorotatory(-) isomers. Both cancel each other. Thus, racemic mixture is optically inactive.

(07) a) Draw cis-trans isomers of hex-3-ene b) what is meant by Levo-rotatory and dextro-rotatory isomers. Give one example?

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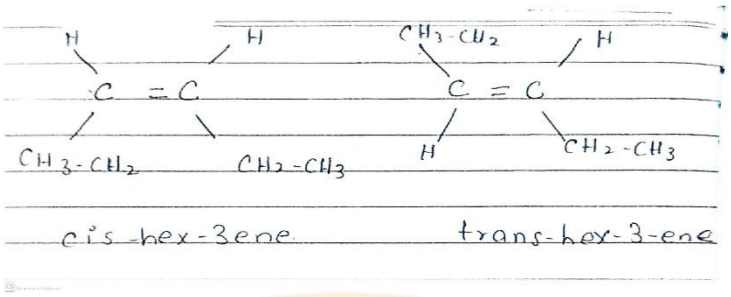
Ans:

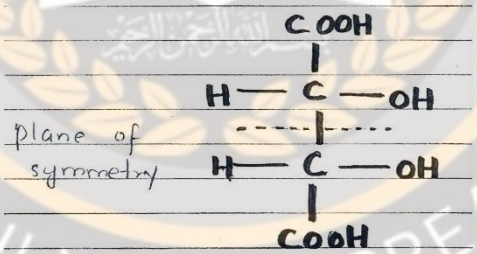


b) Levo-rotatory isomers rotate plane polarized light to left (anticlockwise) while dextrorotatory (+) isomers rotate it to right (clockwise). Example: (+)-limonene occur in lemons and oranges, while (-)-limonene occur in pine, peppermint and spearmint.

(08) a) Why Cis-isomers generally less stable but more reactive than their trans-isomer counterparts? b) How can a molecule be chiral yet remain optically inactive? Give term used for such molecule.

Ans: a) Cis- isomers are less stable because bulky groups are close to each other, causing steric hindrance. They are more reactive because the electron cloud is shifted to one side, making them more open to electrophilic attack. Trans-isomers are more stable as bulky groups are farther apart.

b) A molecule can have chiral center but remains optically inactive if it has internal symmetry that cancels optical rotation. Such molecules are called meso-compounds. Meso-tartaric acid is diastereomer of (+)-tartaric acid. It is optically inactive due to internal plane of symmetry.



(09) a) Why does Cis-isomer have higher boiling point than trans-isomer? b) What is optical resolution and how it is used to separate racemic mixture?

Ans: a) Cis-isomers are polar and have a net dipole moment. This increases intermolecular dipole-dipole attractions. Stronger attraction result in higher boiling point.

b) Optical resolution is a process of separating a racemic mixture into its enantiomers. It uses a chiral resolving agent or technique to form diastereomeric derivatives. These diastereomers have different physical properties and can be separated by crystallization or chromatography.

Chp:8 Hydrocarbons

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(1)(a) How are Aliphatic and aromatic hydrocarbons classified? (b) What was the contribution of Micheal Faraday and Mitscherlich in the discovery of benzene?

Ans: **a) Aliphatic Hydrocarbons:** All hydrocarbons that do not have benzene ring. The word Aliphatic comes from Greek word "Aleiphas" meaning oil.

Aromatic Hydrocarbons (Arenes): Hydrocarbons that contain at least one benzene ring. The word arene has two parts- "ar" means aromatic and "ene" comes from benzene showing unsaturation.

b) Michael Faraday isolated hydrocarbon from oily residue collected in gas pipes of London. He named it “bicarburet of hydrogen”. In 1834, Eilhardt Mitscherlich prepared the same substance. He heated benzoic acid with lime and obtained benzene. He found that empirical formula is CH.

(2) Explain the use of ortho, meta and para in nomenclature of disubstituted benzene?

Ans: The position of substituents on benzene ring can be indicated by use of numbers or Greek words ortho, meta and para. The word "ortho" means meeting, "meta" means beyond and "para" means straight. 1,2 are equivalent to ortho, 1,3 equivalent to meta and 1,4 to para.

(3) a) What is open chain structure of benzene? b) what inspired Kekule to propose cyclic structure of benzene?

Ans: a) After discovery (1825), the structure of benzene molecule confused the 19th-century chemists for forty years. Elemental analysis proved that Benzene has very low hydrogen to carbon (H:C) ratio. Empirical formula "CH" and molecular formula C6H6. Hence benzene shows high degree of unsaturation. Chemists imagined the molecule as an open chain. e.g.

b) One cold night, he had a dream in his laboratory. In his dream, he saw atoms dancing and forming chains. The chains turned into rings. They looked like snakes biting their tails. This gave him an idea. He proposed that benzene has ring structure with six carbon atoms.

(04) a) What are objections to Kekule's structure? b) Why Kekule's structure or model is generally used?

Ans: a) Kekule's structure was accepted for many years. But later it was questioned because of some objections:

- If benzene has double bond like alkenes, then why benzene does not react like alkenes?
- If benzene has three double bonds, why it cannot add three moles of benzene?
- Why benzene favours substitution reaction over addition reaction?
- Resonance exist in more than one resonance form but usually kekules structure is used for convenience.
- It is because it shows the tetravalency of carbon and total number of electrons in benzene.
- it also make it easier to show the movement of electrons in benzene reaction mechanism.

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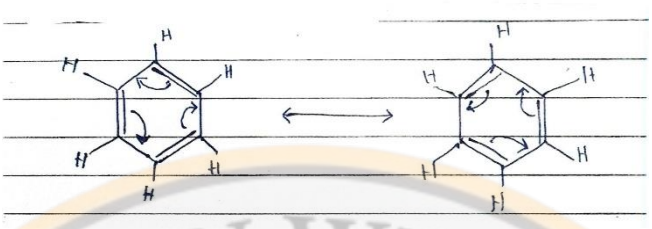
(05) Draw two kekule structures of benzene. How did kekule switch to closed structure benzene molecule although other chemists tried open structure for it? b) Define Resonance? Why is the benzene stable than alkenes?

Ans.Two kekule structure are:



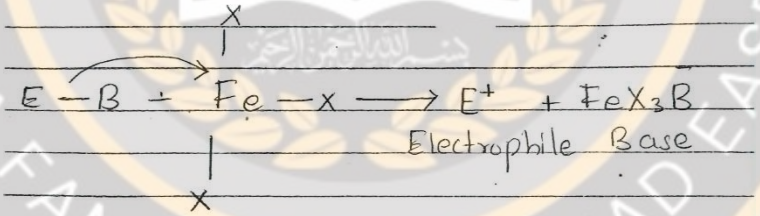
b) Resonance: delocalization of pi electrons over six carbon atoms in a benzene ring. The two kekule structure are only resonance contributing structures—the true benzene is a resonance hybrid of these structure.

(06) a) Why does benzene require a catalyst for electrophilic substitution reactions? b) What is the general mechanism of electrophilic aromatic substitution reaction?

Ans. a) The benzene molecule is in delocalized \pi electrons system far more stable than localized \pi bonds in alkenes. This is why benzene molecule always requires a catalyst for electrophilic substitution reactions.

(b) Generation of Electrophile.

An electrophile is generated first from suitable reagent with help of a lewis acid catalyst, like FeX3.



(07) a) What different products could be obtained by chlorination of toluene? b) what happens when toluene is treated with chlorine gas in light?

Ans a) Halogenation of methylbenzene or alkyl arenes involved electrophilic attack at position 2 and 4. The alkyl substituents are ring activating and increase electron cloud at position 2 and 4 in benzene ring.

(b) Side chain Halogenation:

Chlorine reacts with alkyl side chain of alkylbenzene in presence of UV light or strong sunlight. This follows free radical mechanism. When chlorine gas is passed over boiling methylbenzene in presence of UV light following reaction occurs:

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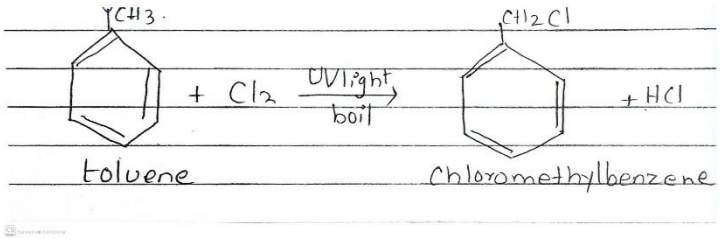
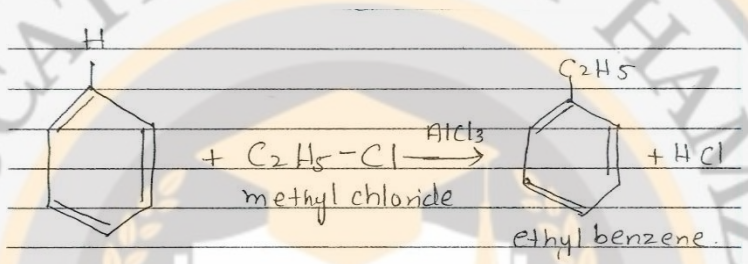


(08) a) How does chloroethane react with benzene molecule? Name the electrophile and end product of this reaction? b) What two roles being played by AlCl3 in Friedal Craft's Alkylation?

Ans:- Chloroethane reacts with benzene by Friedal Crafts alkylation.

AlCl3 is used as catalyst. AlCl3 helps generate ethyl cation ($C2H5^+$) electrophile. The end product of this reaction is ethyl benzene.



(b) Catalyst:

* AlCl3 helps generate electrophile (e.g., $CH3^+$ or $CH3CO^+$) without being consumed.

* AlCl3 carries halogen from chloroalkane then help remove hydrogen from benzene to restore arenium ion and release AlCl3 at end as degenerated ring.

(09) a) Why is aldehyde group termed as 3-directing substituent? b) How 2 and 4-directing substituents donate electrons to benzene ring?

Ans.a) Aldehyde group is 3-directing substituent because it withdraws electrons from benzene ring. This makes ring electron deficient. In arenium ion formed during electrophilic aromatic substitution, ortho and para attacks are destabilized while meta attack is less destabilized.

(b) Ortho/para directing substituents donate electrons to benzene ring by resonance and increase electron density especially at position 2 and 4. This stabilizes arenium ion formed in electrophilic substitution.

(10) a) Explain how is arenium ion formed in electrophilic substitution reactions of benzene? b) State difference b/w substitution reactions of alkanes and benzene?

Ans:- a) The electrophile attacks high electron density of \pi system of benzene. A pair of pi-electrons from ring forms bond with electrophile, breaking aromaticity. The resulting unstable intermediate carbocation is called arenium ion (\sigma complex) which is resonance stabilized.

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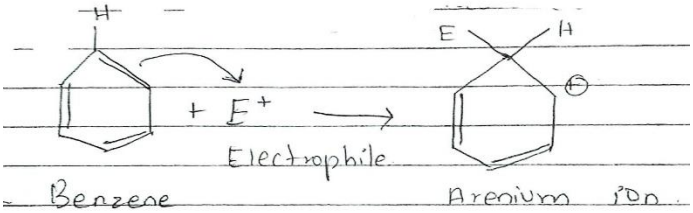

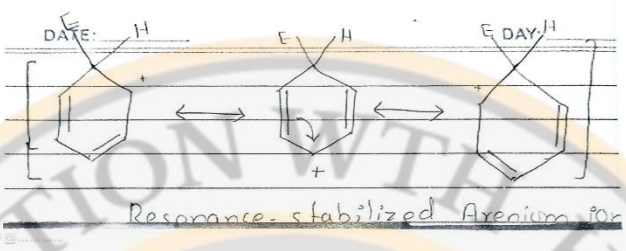

In Alkanes	In Benzene
• Substitution is free radical type.	• Substitution is electrophilic type
• Initiated by UV light.	• Needs Lewis acid catalyst.
• Intermediates are free radicals.	• Intermediate is a resonance stabilized arenium ion.
• less selective and give mixture of product eg:- CH4 + Cl2 →CH3Cl + HCl	• More selective and generally give one product. eg: C6H6 + Cl2 →C6H5Cl + HCl

(11) a) Why Carbon-carbon benzene has lower energy than hypothetical 1,3,5-trien? Why benzene has delocalized double bond? b) The Carbon-carbon bonds are partial double bond. Comment?

Ans:- a) Each carbon atom has an unhybridized 2p orbital. Overlapping sideways, forming a continuous \pi-electron cloud. Hence, the bonds are delocalized instead of alternating double bonds. Resonance delocalization makes benzene more stable, with resonance energy of 152 kJ/mol lower than expected from 1,3,5-cyclohexatriene.

(b) Experimental bond length of benzene (1.39 A°) lies b/w single (1.54 A°}) and double (1.30 A°). Hence all bonds are equivalent partial double bonds due to resonance.

(12) a) Differentiate b/w Electrophilic substitution and addition reactions of benzene? b) Explain why benzene undergoes substitution instead of addition reactions.

Electrophilic Substitution Reaction	Addition Reaction
1- In E.S, one hydrogen atom of benzene is replaced by new substituent.	1- In A.R, other atoms or hydrogen are added across double bonds of benzene.
2- The reaction usually occur under mild condition with a catalyst.	2- The reaction requires harsh condition.
3- Example: Nitration of benzene	3- Example: Hydrogenation of benzene

b) Benzene has delocalized pi-electrons cloud. Breaking it requires harsh conditions. In substitution, one hydrogen is replaced and aromaticity is restored. Addition would break the whole pi-system making benzene unstable.

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(13) a) Differentiate b/w Ortho/Para and Meta directing groups? b) Convert benzene to benzoic acid.

Ortho/Para Directing Group	Meta-Directing Group
1- It donate electrons to benzene ring and increase electron density.	1- This group withdraw electrons from benzene ring and decrease electron density.
2- These groups make ring more reactive toward substitution.	2- These groups make ring less reactive towards substitution.
3- Examples:- -CH3, -OH	3- Examples: -NO2, -CHO

Chp: 09 Halogenoalkanes

(1) a) Define bond polarity? What is leaving group? b) What is nucleophile?

Ans a) Bond Polarity: In a covalent bond, electron density is drawn more towards a more electronegative atom. In alkyl halides, the carbon-halogen (C-X) bond is polarized because the halogen is more electronegative than carbon. Carbon gets a partial positive charge and halogen get partial negative charge. e.g: C^{8+}} – Cl^{8-}}

Leaving Group: The halogen of an alkyl halide is called leaving group. It leaves with an unshared pair of electrons during nucleophilic substitution. Good leaving groups are Cl^-, Br^-, I^- and $HSO4^-$ while poor are OH^-, OR^-, $NH2^-$.

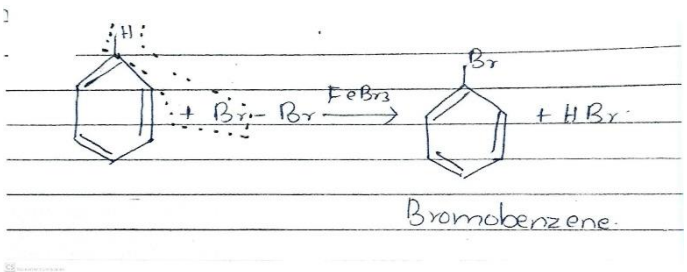
b) A nucleophile is species that has lone pair of electrons and can donate it to an electrophilic carbon of an alkyl halide. It may be neutral or negatively charged. eg:- (OH^-),

(2) a) Differentiate b/w E1 and E2 mechanisms?

E1 Reaction	E2 Reaction
1- It is two step mechanism.	1- It is single step mechanism
2- Rate = k[R-X]	2- Rate = k[R-X][Base]
3- It is a unimolecular reaction.	3- It is bimolecular reaction.
4- Favored in polar protic solvents.	4- Favored in strong basic, aprotic conditions.

(3) a) What is major product when benzene reacts with Br2 in presence of FeBr3? Describe the difference b/w SN1 and SN2 mechanism.

Ans a) The major product is bromobenzene.



SN1	SN2
* It is a two step mechanism.	* SN2 is a one step mechanism.
* First step is slow and second is fast.	* It has only one step and that is slow.

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* It is uni-molecular reaction.	* It is a bi-molecular reaction.
* It is favoured in polar solvents.	* It favoured non-polar solvents.

(04) a) Why do primary halogenoalkanes prefer the SN2 type of mechanism? b) What an intermediate is formed during an SN1 reaction?

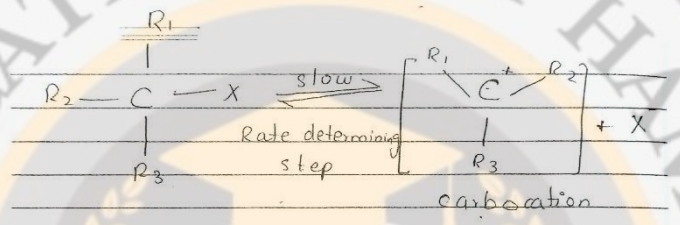
Ans: a) Primary alkyl halides have less steric hindrance. Thus, nucleophile attacks Carbon attached to halogen easily in one step. This gives inversion of configurations.

eg:- $CH3CH2Br + OH^- \rightarrow CH3CH2OH + Br^-$

b) A carbocation is formed during an SN1 reaction.

* The substrate R-X first ionizes reversibly into Carbocation (R^+) and X^- ions.

* This is slow and rate determining step.



(05) a) Is iodine more nucleophilic than Chlorine? b) Halogens are good leaving groups why?

Ans: a) Iodide ion is more nucleophilic than Chloride ion. It is because I^- ion is Large and polarizable while Cl^- ion is comparatively smaller and less polarizable. Hence, I^- ion is more nucleophilic than that of Cl^-.

b) Groups act as leaving groups because they are always capable of accommodating negative charge through high electronegativity or by delocalization. Because halogen atoms have high electronegativities, they form relatively stable ions and act as good leaving groups.

Chp: 10 Hydroxy Compound

(1) Differentiate b/w Alcohols and Phenols.

Alcohols	Phenols
1. In Alcohols, hydroxyl group is attached to a saturated carbon atom. This Carbon is sp3 hybridized.	1. In phenols, the hydroxyl group is attached directly to an aromatic ring (benzene ring).
2. Alcohols are simplest types of organic hydroxyl compounds.	2. Phenols are organic hydroxyl compounds
3. Example: Methanol which is often used as an antifreeze or solvent	3. Example: Phenol reacts with bromine in presence of CCl4 gives mixture of 2- & 4-bromo phenol.

(2) a) How can Acid halide be converted into ester. Write mechanism for reaction? b) How does aniline give phenol on treatment with Nitrous acid?

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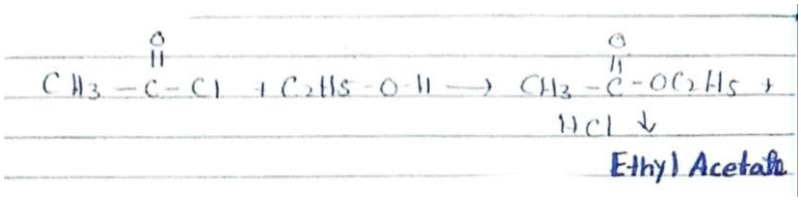
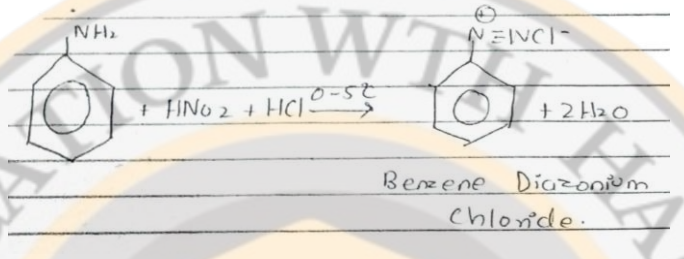
Ans: a) Esters can be prepared directly by reaction of acyl halide with alcohols.

eg: Ethanoyl Chloride reacts immediately with cold ethanol to form ethyl ethanoate.



b) Aniline reacts with nitrous acid in presence of HCl. The reaction must be carried out below 10°C.



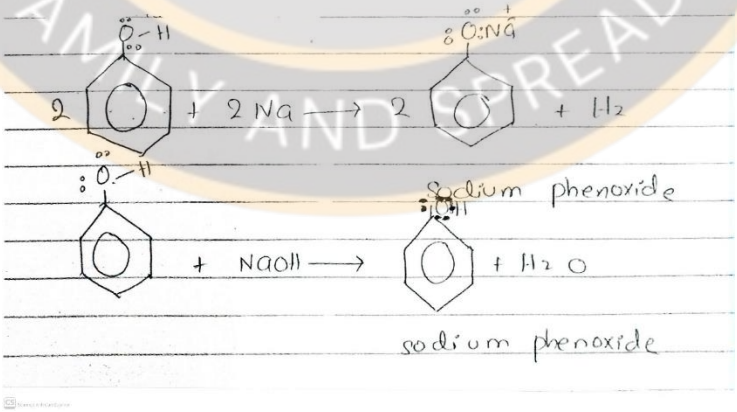
(3) a) Why phenol shows acidic nature? b) Why Ethanol is weaker acid than phenol?

Ans: a) Phenol is stronger acid than H_2O. When phenol loses a proton, it forms Phenoxide ion. The phenoxide ion gains stability through Resonance. The negative charge of phenoxide ion is delocalized into the aromatic ring. This delocalization makes the phenoxide ion stable.

b) Ethanol is a weaker acid than both H_2O and phenol. The conjugate base of ethanol does not have a resonance-stabilized structure. The negative charge is localized in ethoxide. The Ka for ethanol (1.26×10^{-16}}) is smaller than water and phenol.

(04) a) How does phenol react with Na and NaOH? b) How does hydroxyl group influence the electrophilic substitution of phenol?

Ans: a) Phenol is acidic in nature and reacts with sodium or sodium hydroxide to form salts.



b) The main reason is the activating effect of the hydroxyl group in phenol. The hydroxyl group donates electrons into the benzene ring. This donation increases the electron density of the ring, making it more reactive toward electrophilic attack.

(05) a) What is diazonium salt formed when phenyl amine reacts with nitrous acid below 10°C? b) Write chemical equation for reaction of phenol with NaOH?

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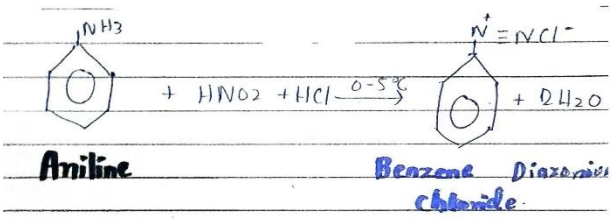
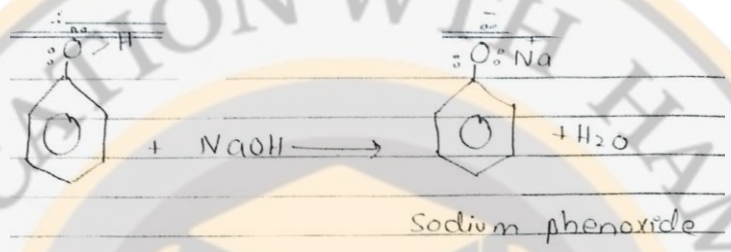
Ans: a) Phenyl amine reacts with HNO_2 in presence of HCl. The reaction must be carried out below 10°C.

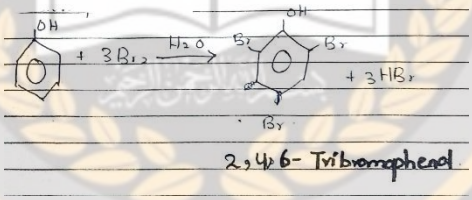


b) Phenol reacts with NaOH because it is acidic. The hydroxyl hydrogen is replaced by Na forming sodium phenoxide.

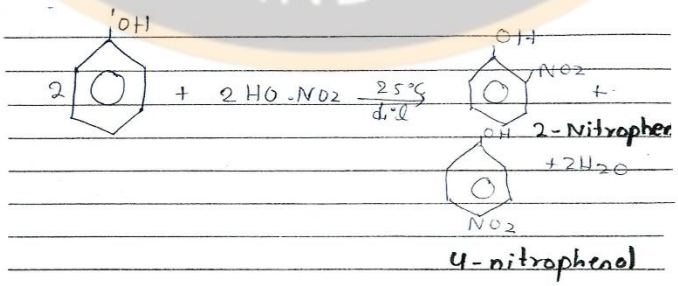


(06) a) What is product when phenol reacts with Bromine water? b) Explain why phenol reacts differently with nitric acid compared to benzene?

Ans: a) When Bromine water (Br2/H2O) is added to an aqueous solution of phenol, the reddish-brown color of Br_2 is decolourized and a white precipitate of 2,4,6-tribromophenol is produced. This reaction is used to identify phenol.



b) Phenol reacts under milder conditions because the -OH group donates electrons into the benzene ring. This increases electron density at 2, 4, 6 positions, making phenol more reactive than benzene. Benzene needs concentrated acids and catalysts, while phenol reacts at room temperature.



(07) Explain why -OH group of phenol is an activating group and ortho-para directing during an electrophilic substitution reactions?

Ans: The -OH is an activating group because it donates resonance electrons to the benzene ring. Negative charge density concentrates at 2, 4, and 6 positions. Electrophiles attack where electron density is highest; hence, substitution occurs mainly at ortho and para positions. This

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is why tribromination and nitration give ortho-para products. It increases the overall reactivity of the ring.

Chp: 11 Carbonyl Compounds

(1) a) Differentiate between Aldehyde and Ketones. b) Differentiate b/w Amides, Esters and Carboxylic Acids.

Aldehyde	Ketones
i) In aldehyde, the carbonyl group is attached to at least one hydrogen atom and one alkyl or aryl group.	i) In ketones, a carbonyl group is attached to two alkyl or aryl groups. |
ii) General formula is RCHO where R can be hydrogen or alkyl/aryl. |	ii) General formula is RCOR where R1 and R2 can be alkyl or aryl.
iii) Example: Acetaldehyde (CH3CHO).	iii) Example: Acetone (CH3COCH3).

b) Amides: These contain a carbonyl group attached to nitrogen (which can be attached to hydrogen or alkyl groups) and an alkyl or aryl group.

* General formula: $RCONR^1R^2$

* Example: Acetamide (CH3CONH2)

Carboxylic Acids	Esters
i) These contain a carbonyl group attached to a hydroxyl group (-OH) and an alkyl or aryl group	i) These contain a carbonyl group attached to an alkoxy group (-OR) and an alkyl or aryl group.
ii) General formula is R-COOH.	ii) General formula is R-COOR
iii) Example: Acetic acid (CH3COOH) or Benzoic acid.	iii) Example: Ethyl acetate (CH3COOCH2CH3).

(02) a) How do electron-withdrawing groups affect the acidity of carboxylic acids and why does this influence the stability of carboxylate ion?

Ans: An electron-withdrawing group increases the acidic strength of carboxylic acids.

Order of acidity for chlorine substituted acetic acids is:

CCl3COOH > CHCl2COOH > CH2ClCOOH

An electron-withdrawing group withdraws electrons from oxygen atom of carboxyl group. So, carboxylate ion is stabilized due to lower charge density on oxygen atom. A stabilized carboxylate ion has less chances to combine with an H^+ ion. Thus, the acid behaves stronger.

(03) a) How do electron-donating groups decrease acidity of Carboxylic Acids? b) Write reaction involving H atom of carboxyl group?

Ans: An electron donating group decreases acidic strength of carboxylic Acids. e.g: alkyl group.

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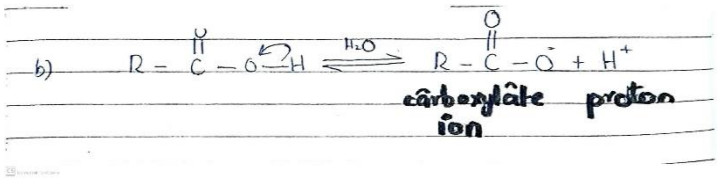
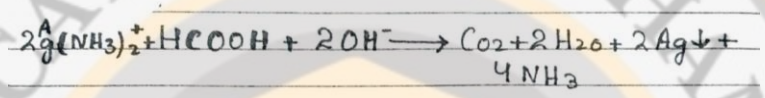
An Alkyl group pushes negative charge towards the carboxylate ion. This Increases its tendency to accept an H^+ ion. This negative charge also strengthens the O-H bond in -COOH group. Thus, it reduces release of H^+ ions.



(04) a) What is Oxidation reactions? b) Write down the reactions of Acyl Chloride?

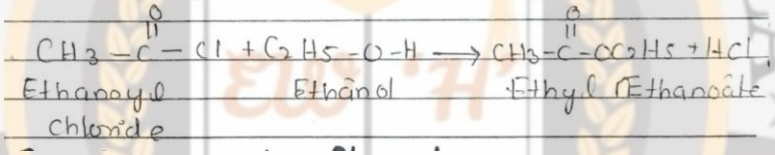
Ans:- a) Some Carboxylic Acids Like methanoic acid and ethanedioic acid can be further oxidized by some oxidizing agents.

Example: Methanoic acid is also oxidized to CO2 and H2O by Tollen's reagent with formation of silver mirror.

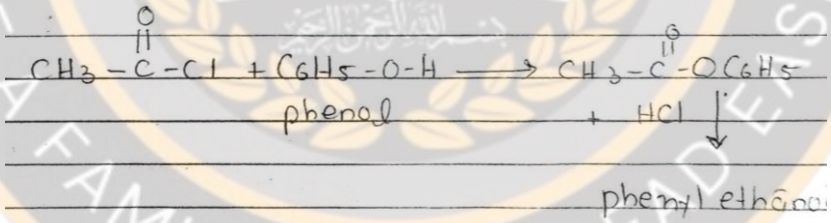


b) Reactions of Acyl Chlorides

Reaction with Alcohol:



Reaction with Phenols:



(05) Why phenylchlorides does not give nucleophilic Substitution Reactions?

Ans: The nitrogen's lone pair is delocalized into benzene ring, making it less available for nucleophilic attack. The ring is electron rich due to resonance repelling nucleophiles. The benzene ring aromaticity is prioritized, resisting substitution.

(06) Explain why a) methanoic acid can be Oxidized while other carboxylic Acids cannot. b) What products are formed when Ethanedioic acid is oxidized with warm acidified KMnO4?

Ans a) Methanoic acid has a -CHO group along with -COOH (H-C(=O)-OH). It behaves like an aldehyde and can be oxidized. Other Carboxylic Acids do not have this -CHO group, so they cannot be oxidized.

(b) Ethanedioic acid is Oxidized to CO2 by warm potassium manganate (VII) solution acidified with dilute H2SO4.

(COOH)2 + 2KMnO4 + 3H2SO4 → 10CO2 + 2MnSO4 + K2SO4 + 8H2O

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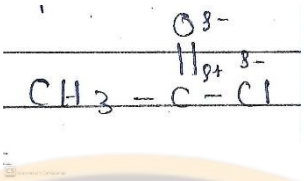
(07) a) what is difference in reactivity b/w acyl chlorides and alkyl chlorides? b) Explain the influences on acidity of chlorine of Carboxylic acids?

Ans a) Acyl chlorides hydrolyze most readily because carbonyl carbon is highly electron deficient.



In alkyl chlorides, carbon is is attached with electronegative atom Cl so it is less reactive. Hydrolysis of alkyl chlorides requires a stronger nucleophile than water.

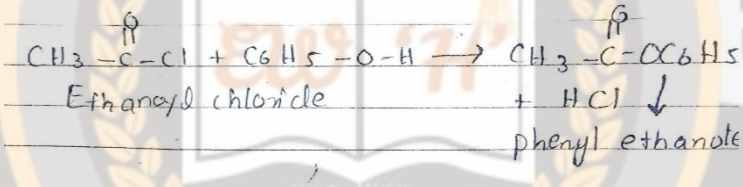
$CH3-CH2^{\delta+}-Cl^{\delta-}$

b) An electron-withdrawing group increases acid strength. The Chlorine atom pulls electrons from oxygen of -COOH and stabilizes the carboxylate ions. So, Acidity increases.

Order is:- CCl3COOH > CHCl2COOH > CH2ClCOOH

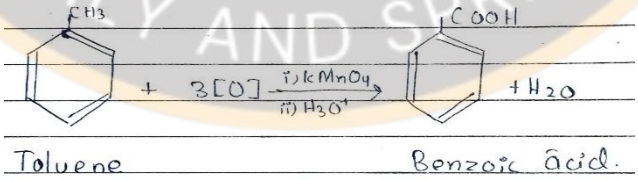
(08) a) What products results from the reaction of acyl chlorides with phenol?

Ans:- Acyl chloride react with phenol at room temperature to produce an ester.



(09) Methyl benzene can be oxidized to benzoic acid using hot alkaline KMnO4 and then dilute acid?

Ans:- Alkaline KMnO4 acts as strong oxidizing agent. It oxidizes side chain of alkyl benzene into -COOH group. The Reaction mixture is acidified with dilute acid to convert salt into benzoic acid.



(10) a) Compare the acidity of Mono, di-, and tri-chloroacetic acid. Explain the trend cleanly? b) Explain how ethanoyl chloride reacts with methylamine to form N-methylethanamide?

Ans a): Three chlorines withdraw electrons more strongly than one. This stabilizes COO^- in trichloroacetic acid. Dichloroacetic acid is less stabilized but stronger than mono. Mono has least electron withdrawal.

So, Order is: CCl3COOH > CHCl2COOH > CH2ClCOOH

Hence, acidity increases with more Chlorine atoms.

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b) Methylamine has lone pair on nitrogen. It attacks carbonyl carbon of ethanoyl chloride. A tetrahedral intermediate forms. Chloride leaves forming amide. HCL produced react with more amine.Ammonium salt is produced as by-product .

$$CH_3-\overset{O}{\overset{\|}{C}}-Cl + NH_3 \longrightarrow CH_3-\overset{O}{\overset{\|}{C}}-NH_2 + HCl$$

Ethanoyl chloride — Ethanamide

$$HCl + NH_3 \longrightarrow NH_4Cl$$

Chp 12: Nitrogen Compounds

(01) How Amines are classified? Give Examples?

Ans:- **Primary Amines**: These are Amines in which N-atom is attached with one alkyl group.

e.g: Ethyl Amine (CH3-CH2-NH2)

Secondary Amines: These are Amines in which N-atoms is attached with two alkyl groups.

e.g:- $CH_3-\overset{CH_3}{\overset{|}{N}}-N$

Diethyl amine (sec-amine)

Tertiary Amines: These are Amines in which N-atom is attached with three alkyl groups.

$CH_3-\overset{CH_3}{\overset{|}{N}}-CH_3$

Triethyl amine (ter-amine)

(02) What types of compounds can be reduced to prepare amines, and which reducing agents are commonly used method?

Ans:- Compounds containing nitrogens such as Nitriles and Amides are reduced to Amino group by various reducing Agents.

a) Reduction of Nitriles: Reduction of alkyl or aryl nitriles gives pri-amines. This reduction is carried out by reducing agent LiAlH4 or sodium in ethanol.

$$R-C\equiv N + 4[H] \xrightarrow{LiAlH_4} R-CH_2-NH_2$$

Nitrile — Amine

b) Catalytic Reduction of Nitriles: Reduction reaction with hydrogen gas in presence of nickel catalyst give primary amines.

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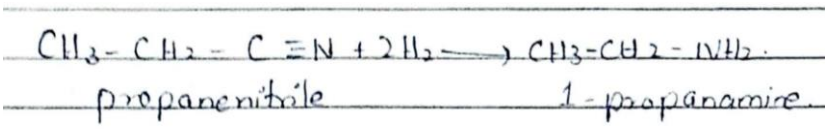
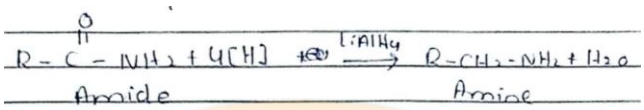


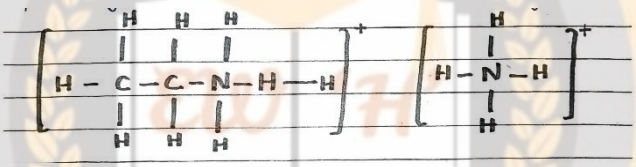
c) Reduction of Amides: When an amide is treated with reducing agent LiAlH_4, -CO group of amide is reduced to -CH2 and primary amine is formed.



(03) a) How does lone pair of nitrogen on amine make the aqueous solutions alkaline like Ammonia? b) Why ethyl amine is stronger base than ammonia aqueous solution Explain?

Ans a) The lone pair of electrons on nitrogen atom makes amines basic in nature. The lone pair on nitrogen in amines behaves in the same way as in ammonia. When an amine is dissolved in H2O, Lone pair accepts protons from the water molecules to produce alkyl ammonium ions and hydroxide ions. The formation of hydroxide ions makes aqueous solutions of amines alkaline.

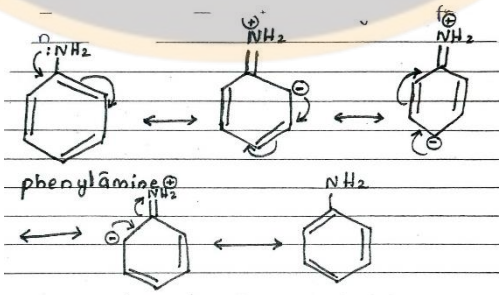
b) Ethylamine and Ammonia have the following structure after accepting H^+ ion.



An ethyl group is an electron-donating group. This group increases electron density on nitrogen. As a result, the lone pair of electrons on nitrogen becomes more electron-rich and attractive to hydrogen ions (H^+).

(04) a) Phenylamine (aniline) is much weaker base relative than ammonia? b) Describe the relative basicities of aqueous ammonia, ethyl amine and phenylamine?

Ans a) Phenylamine is an aromatic primary amine. In this, the lone pair of electrons on nitrogen atom take place resonance with benzene ring and became delocalized. So, lone pair of electrons cannot easily accept a proton (H^+).



Thus, negative charge around nitrogen atom in phenylamine is much less than in ammonia. The order of basic strength of these compounds is:

Ethylamine > Ammonia > Phenylamine

b) The relative strengths of weak bases are determined from their pKb values.

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Base	pKb
Ammonia	4.75
Ethyl amine	3.27
Phenylamine	9.38

* Lower value of pKb, the stronger the base.

* Thus, ethylamine is a stronger base than ammonia, which in turns is stronger than phenylamine.

Two factors determine the basic strength of substrate.

(i) the ease with which its lone pair can accept the H^+ ion.

(ii) stability of ion form after accepting H^+ ion.

(05) Explain the reactions of the hydrolysis of Nitriles?

Ans: **Hydrolysis of Nitriles:**

Hydrolysis of nitriles can occur with both dilute acid or dilute alkali.

Acid hydrolysis of Nitriles:

When a nitrile is heated with dilute acid such as dilute HCl or dilute Sulphuric acid, carboxylic acids are formed.

$CH_3CN(aq) + 2H_2O(l) + HCl(aq) \rightarrow CH_3COOH + NH_4Cl$ **(Ethanoic Acid)**

Alkaline Hydrolysis of Nitriles:

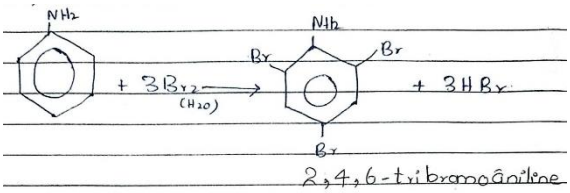
When a nitrile is heated with a dilute sodium hydroxide solution, a salt of carboxylic Acid is produced. This salt on acidification with dilute HCl gives free carboxylic Acid.

$CH_3CN(aq) + H_2O(l) + NaOH(aq) \rightarrow CH_3COONa(aq) + NH_3$

$CH_3COONa(aq) + HCl(aq) \rightarrow CH_3COOH(aq) + NaCl(aq)$

(06) Write down the reaction of phenyl Amine with bromine water? Also write the prepration of Diazonium salts?

Ans: **a) Reaction with Bromine Water:**



A solution of phenyl amine in HCl is cooled below 5°C. The sodium nitrite or potassium nitrite solution is also cooled in ice. This ice cold solution added to phenyl amine solution produces solution that contains diazonium chloride. Phenyl diazonium chloride is also called Benzene Diazonium Chloride.

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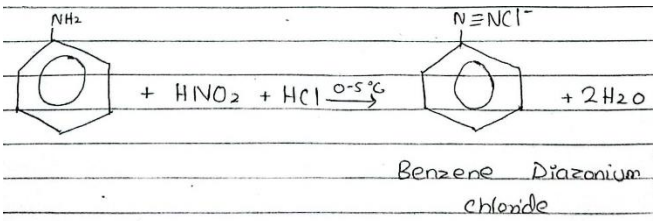

(07) a) How may phenol be converted into azo dyes? b) Why are this group makes them strongly colored?

Ans. a) Phenol is dissolved in NaOH solution and its ice cold solution mixed with ice cold solution of diazonium chloride. A coupling reaction take place producing yellow-orange compound. These compound are known as azo-compounds and one of simplest compound called azo-dyes.

b) The -N=N- unit is called an azo group. Compounds containing this functional group are called Azo dyes. In azo dyes two benzene rings are linked by nitrogen bridge (-N=N-). This extends conjugation and make them strongly colored. Azo-dyes make up more than 60% of total Industrial dyes. About 70 % of total dyes of industries are azo dyes.

(08) How does amide products differ when acyl chlorides react with ammonia compared to when they react with primary amines?

Ans:- **By the Reaction of Acyl chloride with Ammonia:**

Acyl halides react with ammonia at room temperature to produce amides. The hydrogen chloride produced reacts with excess ammonia to give ammonium chloride.

$CH_3COCl + NH_3 \rightarrow CH_3CONH_2 + HCl$

ETHANAMIDE

$HCl + NH_3 \rightarrow NH_4Cl$

By the Reaction of Acyl chloride with Primary Amines:

Primary amines react with ammonia in similar way at room temperature to produce N-substituted amides. The excess amine reacts with hydrogen chloride to give ammonium salt.

$CH_3COCl + CH_3NH_2 \rightarrow CH_3CONHCH_3 + HCl$

(09) a) Amides are much weaker bases than amines? b) Write physical properties of Amino acids?

Ans: a) The Amid group has carbonyl group. This group is highly electronegative. It pulls electrons towards it by resonance as well as inductive effect. So, its NH_2 group has not lone pair of electrons around it. So it has little ability to accept proton. In amines, lone pair of electrons on nitrogen is not delocalized. Moreover, alkyl group increases electron density on N atoms. So its NH_2 group has strong ability to accept proton H^+.

b) Physical Properties of Amino acids:

Amino acids has both basic amine group and an acidic carboxylic acid group. There is an internal transfer of hydrogen ion from -COOH group to -NH_2 group. This forms an ion with both negative charge and positive charge called Zwitterion.

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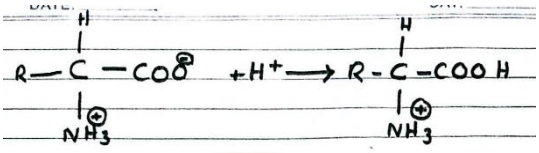
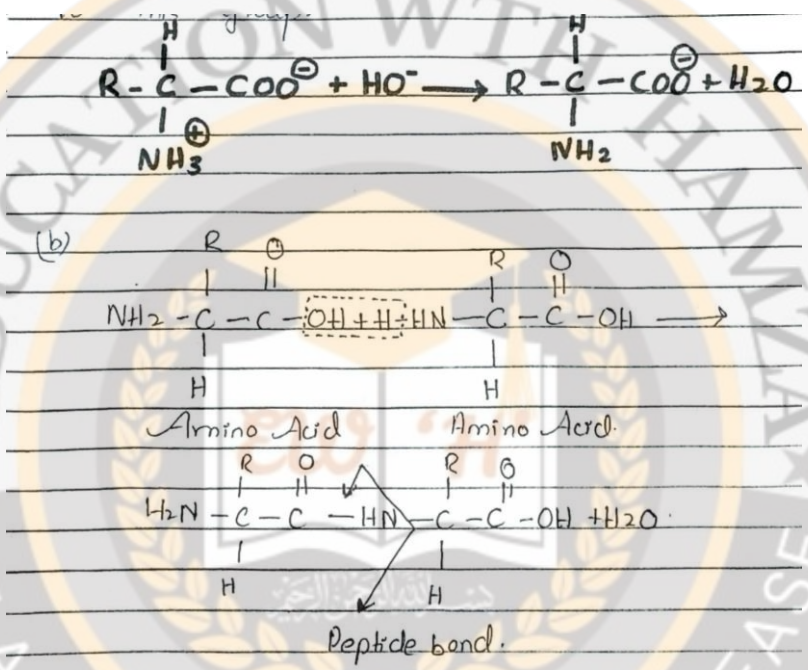
(10) a) How does aminoacids behaves as an acid and base? b) Write down the reaction for the formation of peptide bond?

Ans a) When an acid is added to an amino acid, the Carboxylate ion accept proton. Therefore, this basic character is due to this group.



When an alkali is added to amino-acid, $NH3^+$ group releases proton and therefore acidic characters is due to this group.



(11) a) Compare the reduction of nitriles and Amides using LiAlH4 b) Analyze the significance of diazonium salt in synthesis of biochemistry?

Ans: nitriles reduced with LiAlH4 gives primary amines. R-C≡N →R-CH2-NH2. Amides reduced with LiAlH4 also gives pri amines: CH3CONH2 → CH3CH2NH2

In both cases , carbon chain remain intact.

b) diazonium salt are very important intermediate. They react with H2O to form p-henols. They also couple with phenol or phenylamine to form azo compound.

Azo dyes are strongly coloured and make up more than 60% of oil dyes. About 70% of all dyes are azo compound.

12) Why is hydrolysis of nitriles considered as two step process? B) why secondary amine are more basic than tertiary amine in aqueous solution?

Ans a) In first step nitrile is converted to amide. In second step amide further hydrolyses to give carboxylic acids. So two separate step occur before acid is obtained.

These steps require heating with dilute acid and base. That’s why nitriles hydrolysis is slower but gives complete conversion.

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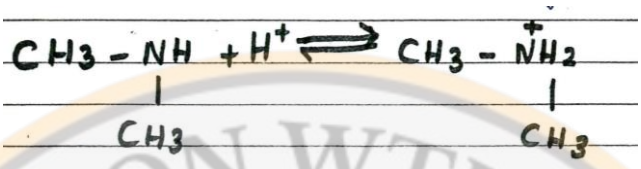
R-C ≡N → R-CONH2 → R-COOH

b) The basicity depends upon:

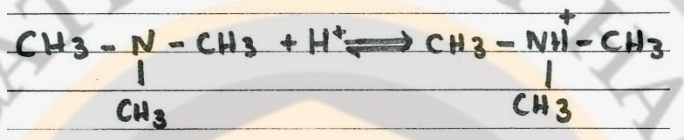
* Availability of electron pair

* The stability of corresponding cation

The cation formed by secondary amine:



The cation formed by tertiary Amine:



The cation of tertiary amine has one hydrogen atoms attached to N can form one hydrogen bond. The cation of sec amine stabilized in aq. solution by hydrogen bonding with more water molecules. Thus, secondary amines are stronger bases than tertiary amines in aq solution.

Chp 13: Polymers

(1) a) Predict the type of polymerization for reaction for following:

i) CH3-CH=CH2

ii) HO-A-OH + HOOC-B-COOH

b) A polymer is made from monomers benzene 1,4-diacyl chloride and benzene 1,4 diamine. Draw its repeat units?

Ans:- **a) i) Propylene CH3-CH=CH2**

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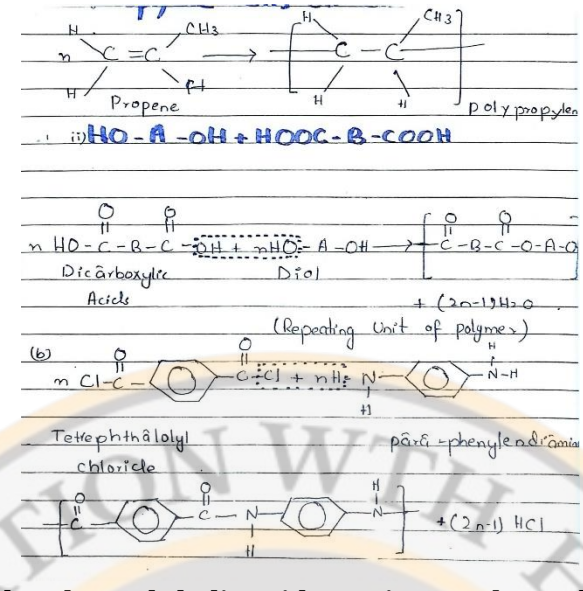
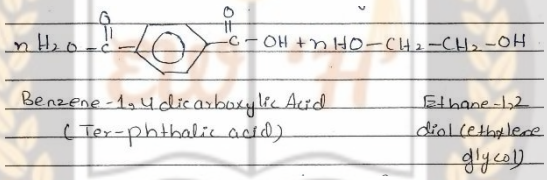
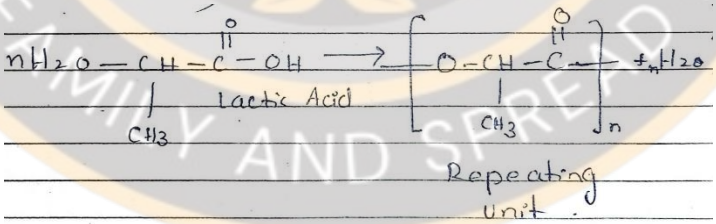


(12) a) Ethylene glycol and ter-phthalic acid reaction produce which polymer? To which class this polymer belongs to? b) Write the reaction for the polymerization of Hydrocarboxylic acids?

Ans: a) The most common polyester is called terylene. It is also called polyethylene terephthalate. Its monomers, ethane-1,2-diol, and benzene-1,4-dicarboxylic Acid.



b) Hydroxy carboxylic Acids also forms polymers. e.g:- polylactic acid. It is made from monomer, 2-hydroxypropanoic acid (lactic Acid).



(3) a) Which type of polymers are biodegradable? Give example. b) What are long term effects of non-biodegradable polymers?

Ans a) Polymers that can be decomposed by the action of microorganisms. e.g:- polyamides & polyesters.

b) Non-biodegradable polymers accumulate in the environment. These do not decompose naturally in presence of sunlight, H2O and air. They can produce highly toxic pollutants. These pollutants enter water and cause water pollution.

(4) a) Describe the chemical reaction b/w an amine and acyl chloride to form an amide? b) Identify the raw material available in pakistan for chemical industry

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Ans a) When a primary amine reacts with an acyl chloride the amine group combines with the -OH group of acyl chloride. This results in the formation of an amide linkage (-CO-NH-) with elimination of HCl as by-product.

$$R\text{-}COCl + R'\text{-}NH_2 \rightarrow R\text{-}CONHR' + HCl$$

b) The basic raw materials are fossil fuels (coal, natural gas, oil), salt, limestone, sulphur, air, water. There are also specialized materials like phosphates, flourides. The country has deposits of gold, chromite, mineral coals, copper, salt and bauxite.

(05) a) Differentiate b/w Condensation and Addition Polymerization? b) How can polyesters and polyamides be degraded?

Addition Polymerization	Condensation Polymerization
i) Monomers add together without losing any atom or molecule.	i) Monomers join together with elimination of small molecules e.g - H_2O.
ii) The polymer has an exact multiple of molecular formula of monomer.	ii) The polymer has molecular formula that is slightly less than total of monomer because of loss of small molecules.
iii) Example: polystyrene, Polyethylene.	iii) Example: nylon-6,6, polyester, Bakelite.

b) Polyesters can be hydrolyzed by Acids or bases. Acid hydrolysis gives diol and dicarboxylic acid, while alkaline hydrolysis gives diol and carboxylic Acid salt. Polyamides also undergo hydrolysis under acidic or basic conditions. In acids they give dicarboxylic acids and ammonium ion, while in base, they give dicarboxylic Acid salt and diamine. That is why polyesters and polyamides are considered as biodegradable polymers.

(06) a) What are the properties of nylon that make it suitable for textiles? b) Describe an application of polymers in biomedical science?

Ans a) Nylon-6,6 is synthetic fiber that is strong and elastic. Its high tensile strength allows it to be drawn into thin fibers without breaking. It also resists abrasion, making it durable for long term use. It is used in clothing, carpets, fishing lines, ropes and Brush bristles.

b) Polymers are widely used in biomedical science for making artificial organs. Mechanical artificial organs are made from non-living polymers such as plastics. They are used in medical devices, artificial body parts and educational tools. This helps the doctors treat patients and also train medical students.

(07a) Why does alkaline hydrolysis of polyester give salt of carboxylic Acid instead of free acid? (b) Why is recycling of plastic necessary even though they are chemically inert?

Ans:- a) During Alkaline hydrolysis the base reacts with Carboxylic Acid groups. It converts them into sodium salt. This reaction is called saponification in case of esters. Because salt is formed and more stable, free acids cannot reform until acid is added. Thus products are diol and Carboxylic Acid salts.

b) Chemically inert plastics do not decay naturally. They remain unchanged for many years in landfills and oceans. Burning them produces toxic gases like dioxins. Recycling melts and reshapes old plastics into new products. This reduces environmental pollution and conserve raw materials like petroleum.

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(8) a) Why does UV light degrade polyethylene but not metals or glass?

Ans: Polyethylene has weak C-C single bonds. That can absorb UV light and break. This process is called Photo-degradation; it breaks long chains into smaller fragments. The plastic becomes brittle and loses strength. Metals have strong metallic bonds and glass has a covalent network structure. They do not absorb UV light the same way. Thus, UV light mainly degrades plastics like polyethylene and polypropylene.

Chp: 15 Biochemistry

(1) a) Define Carbohydrates? b) write down function of carbohydrates?

Ans a) Carbohydrates are most abundant class of organic compounds. They are macromolecules made up of elements carbon, hydrogen, oxygen. General formula Cx(H2O)y.

b) Functions of Carbohydrates:

i) They provide energy.

ii) They provide structural support to body.

iii) They support to Organism's growth and development.

(2) How Carbohydrates are classified? Describe three properties of each?

Ans:- Monosaccharides, Oligosaccharides, Polysaccharides.

*** Monosaccharides:** are simplest type of carbohydrates. They cannot be hydrolyzed into smaller carbohydrates units.

* Their general formula (CH2O)n.

* They are classified into trioses, tetroses, pentoses and hexoses depending upon number of Carbon atoms. They are soluble in water and have sweet taste.

*** Oligosaccharides:** that produces 2 to 10 monosaccharide molecules on hydrolysis. Oligosaccharides are further classified into disaccharides, trisaccharides.

* They are soluble in H2O.

* They have sweet taste.

* They are white crystalline solids.

*** Polysaccharides:** that produces hundred to thousands of monosaccharide (simple sugar) units on hydrolysis.

* They are amorphous solids.

* They are tasteless and insoluble in H_{2}O.

* They are non-reducing in nature.

(03) a) Define proteins? b) Based on structure how they are classified?

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Ans a) Proteins are Nitrogenous, high molecular weight polymers. The building blocks of all proteins are amino acids. Therefore proteins produce amino acids upon hydrolysis.

b) Based on structure:

*** Primary structure of Proteins:** The sequences of amino acids in peptide chain. Amino acids are linked together by peptide bonds. The arrangement of amino acids in chains define primary structure.

*** Secondary structure of Proteins:** Peptide chains may be spiral shape or zig-zag pattern. This coiling arrangement of polypeptide is called secondary structure of proteins. The secondary structure is stabilized by hydrogen bonds. e.g:- collagen.

*** Tertiary Structure of proteins:** The twisting or folding of polypeptide chains represents the tertiary structure. e.g myoglobin.

*** Quaternary Structure of Proteins:** They are formed by arrangement of multiple folded or coiled protein molecules in multi-subunit complex. Different bonding interactions, including hydrogen bond, salt bridges and disulfide bonds hold the chains in specific geometry. e.g.. Hemoglobin.

(4) a) How proteins are classified on the basis of constitution? b) Differentiate b/w fibrous and Globular proteins?

Ans a) Proteins are classified into:

*** Simple proteins:** Proteins that on complete hydrolysis, produce only amino acids. Example: Albumins, Globulins.

*** Derived Proteins:** Proteins are formed by partial hydrolysis of simple or conjugated proteins through acid, bases, enzymes. Example peptides, proteans.

*** Conjugated Proteins**: Proteins composed of simple proteins (apoprotein) and non-protein group (prosthetic group). Example:- Glycoproteins, Hemoproteins.

Fibrous Proteins	Globular Protein
Made up of Elongated fibrous polypeptide chains.	Having spherical globule like shape upon folding.
They form fibers or large sheets.	Water soluble in acid, alcohol.
Connective tissue contains Collagen e.g: myosins.	e.g: Albumin.

(05) How proteins are classified on basis of function?

Ans:- Proteins are classified into functional categories:

*** Enzymes:** Proteins that facilitate biochemical reactions. e.g: Lactase break down lactose into glucose and galactose.

*** Structural Proteins:** Proteins providing support and shape to cells and tissue. e.g: collagen provides strength and elasticity to connective tissues, liver and spleen.

*** Storage Proteins:** Proteins that store nutrients and ions. e.g Ferritin stores iron. Casein store calcium in milk.

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*** Defensive Proteins:** Proteins that protect against pathogens. e.g: Antibodies bind to and neutralize bacteria and viruses.

(06) Differentiate b/w Induced fit model and Lock and key model?

Lock and Key Model	Induced fit Model
Emil Fischer (1894) proposed the lock and key model of enzymes.	Daniel Koshland (1958), American biologist, proposed this model.
In this model, enzymes and substrates have specific shapes that fits perfectly, explaining enzyme specificity.	According to this model, active sites of enzymes are not rigid. Active site molds itself into desired shape.

7) how does the enzyme effect the role of enzyme reaction? How does the pH effect the role of chemical reaction?

Ans: a) Enzymes are very sensitive to their environment any factor that Changes the shape of an enzyme. Enzyme work faster at a specific optimum. Animal enzymes work best at 37°C. Plant enzymes at 60°C. Increasing temperature adds kinetic energy. It speeds up atoms vibrate excessively. If temp is too high, enzyme structure which is the globular structure is lost and denaturation occurs.

b) PH: Enzymes work with specific optimum pH range. Small changes in pH can slow or stop enzyme activity. Trypsin works in small intestine at alkaline PH 8-9.

(8) a) How inhibitors play role in Enzyme-catalyzed Reactions. b) Differentiate b/w Competitive and Non-Competitive inhibitors?

Ans: a) Inhibitors regulate enzyme activity by modulating enzyme-catalyzed rxs. It involves the competitive inhibitor that is similar to substrate. It competes with substrate for enzyme's active site. This prevents substrate from attaching.

Competitive Inhibitors	Non-Competitive Inhibitors
It prevents the substrate from binding by physically blocking the active site.	It changes the shape of the enzyme so active site no longer fits the substrate.
The maximum reaction rate remains the same but a higher substrate concentration is required.	The maximum reaction rate decreases because fewer active enzymes molecules are available for reactions.
E.g: Malonate is competitive inhibitor of Succinate Dehydrogenase.	E.g: Cyanide is non-competitive inhibitor of cytochrome oxidase.

(09) a) What are enzymes? why are they called Biocatalyst b) How does enzyme work?

Ans: a) Enzymes are proteins that speed up biochemical rxs in living cells. They are highly specific and act as mild temperature and pH. They are called Biocatalysts because they work like catalysts but are made by living organisms. e.g: Amylase convert starch into maltose quickly without being used up.

b) Enzymes work by binding with substrate to form an enzyme-substrate complex. This lower the activation energy and convert substrate into product. After reaction enzyme is released unchanged and can be reused. e.g Sucrase breaks into glucose and fructose.

(10) a) Define lipids? b) Describe its three classes with properties?

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Ans: a) Lipids are three naturally occurring heterogeneous organic compounds found in animals and plants. They are soluble in water. They include fats, oils, hormones, waxes.

b) 1. Triglycerides: 1. Triglycerides are found in oils and butters and adipose tissue.

2. Fats and oils are made of kinds of molecules glycerol and three fatty acids.

3. Since they contains two fatty acids so called triglycerides.

2. Phospholipids: 1. they are the esters of glycerol, two fatty acids and one phosphate group.

2. They are integral part of cell membrane.

3. It provide support and strength to plants cell membrane.

3. Waxes: 1. Consist of long chain fatty acids esterified to long chains alcohol.

2. They often used for protective cover in plants and Animals.

3. E.g: pinene, camphor, menthol.

(11) a) Differentiate b/w RNA and DNA? b) Write the names of structural components of DNA and RNA?

DNA	RNA
It is usually double stranded and form a stable double helix structure.	It is usually single stranded and can fold into various shapes.
DNA stores genetic information and passes it to next generation	It help to Protein synthesis and perform various function in cell.
It is found in cell Nucleus.	It is found in nucleus as well as in Cytoplasm.

b) DNA and RNA made of nucleotides. Each nucleotide has three parts:

* A pentose sugar (Deoxyribose in DNA) (Ribose in RNA).

* A nitrogenous base (A, G, C, T in DNA) (A, G, C, U in RNA).

* A phosphate group links nucleotides together.

(12) a Differentiate b/w saturated and Un-saturated fatty Acids? b) What is function of Hemoglobin?

Saturated Fatty Acids	Unsaturated Fatty Acids
Having no double bonds b/w carbon atoms of hydrocarbon chain.	Having one or more carbon-carbon double bonds present in hydrocarbon chain.
They are commonly found in butter, ghee and Animals fats	Commonly in vegetables oils like sunflower oil
They are usually solid at room temperature because of their tight packing.	They are liquid at room temp because of loose packing.

b) Hemoglobin is essential globular protein. Found inside red blood cells and function for respiration. Its main function is to carry oxygen from lungs to all body tissues and carry CO_2 back to lungs for exhalation.

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12) a) What role do antibodies play in immune system? b) What is optimal pH for human enzyme?

Ans: a) Antibodies are special proteins made by B-lymphocytes. They recognize and attach to foreign antigens present in bacterias and viruses. This helps to neutralize pathogens.

b) Most human enzymes work at pH 7, which is neutral. e.g: Salivary amylase acts at pH 7. The enzymes have different pH optima. Pepsin in stomach works at very acidic pH 2. If pH become very different from optimum, enzyme activity decreases.

(13) Identify sources of minerals such as iron, calcium, phosphorous and zinc?

Ans: **Iron** is found in red meats liver and green leafy vegetables. Calcium is obtained from milk, cheese and other dairy products. **Phosphorous** is present in nuts. **Zinc** is found in meat, fish, eggs, cereals, beans, seafood. These minerals are essential for blood formation, bone strength, energy metabolism and immune function.

(14) How does insulin regulate blood glucose levels?

Ans: Insulin is hormonal protein secreted by B-cells of pancreases when blood glucose level rises after a meal. It act as signal for body cells to take up glucose from bloodstream. Insulin promotes conversion of excess glucose into glycogen in liver and muscles which helps store glucose for further energy needs.

Chp: 11 Quantitative Analysis

(1) How you calculate average atomic mass of an element using relative abundance of all isotopes?

Ans: The average atomic mass of an element is determined by formula.

Relative mass= $\frac{\sum(mass\ no) \times isotope\ abundance}{100}$

Example: The study of Boron (Z=5). Spectrum of isotopes tells relative abundances with mass no of isotopes.

* Mass no of first isotope of B = 10. Relative abundance of B-10 = 19.90%.

* Mass no of 2nd isotope of Boron = 11. Relative abundance = 80.10%.

Average Atomic mass of Boron:

$= \frac{(10 \times 19.90) + (11 \times 80.10)}{100} = 10.801$ ams.

(2) a) How no of carbon atoms in a molecule can be calculated by using intensities of peaks in mass spectrum? b) How you can identify the presence of chlorine and bromine in molecule using mass spectrum?

Ans: a) The M+1 peak arises due to presence of 13C Isotopes in a molecule. Its relative intensity helps calculate number of carbon atoms using formula.

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No of atoms= $\frac{100 \times abundance\ of\ M{+}1\ peak}{1.1 \times abundance\ of\ M^{+} peak}$

example: If Abundance are M^+ = 27.32% and M+1 = 2.10% then:

No of carbon atoms= $\frac{100 \times 2.1}{1.1 \times 27.32}$ = 7

This method is highly useful in deducing molecular formula from spectrum.

b) Chlorine has two isotopes: Cl-35 (75.77%) and Cl-37 (24.23%) giving a molecular ion peak at m/z = 35 and an m+2 peak at m/z = 37. The m+2 peak is one third the height of molecular ion peak due to one third abundance of Cl-37 isotope.

Bromine has almost equal isotopes: Br-79 (50.5%) and Br-81 (49.5%). Thus, it produces an molecular ion peak and M+2 peak of nearly equal height.

Hence Chlorine M+ peak and M+2 peak with 3:1 and Bromine M+ peak and M+2 peak with almost height confirms bromine.

(3) a) Explain peaks and how they help in mass spectrometry analysis. b) How can mass spectrometry help in studying reaction mechanism?

Ans: a) The base peak represents the most abundant and stable ion in mass spectrum and is assigned a relative abundance of 100.

Example: In n-pentane, propyl cation ($CH3CH2CH2^+$) give the base peak at m/z = 43. Other peaks like methyl cation m/z = 15. Base peak help identify most stable fragments providing information about molecular structure.

b) Mass spectrometry can detect molecular fragment cations and intermediate cations formed during reactions. By analyzing fragments and masses, chemists can determine which bonds break and how molecules rearrange.

e.g: Alcohols may show M-18, M-17 peaks due to hydrogen or OH loss. This helps in understanding stepwise process of chemical reaction at molecular level.

(1) a) Why sometimes molecular ion peak is smaller than fragments peaks and why does this tell us? b) What is difference between M+1, M+2 peaks and what information do they provide?

Ans a) The molecular ion is often less abundant because it can easily undergo fragmentation. Certain fragment peaks may be higher due to the stability of molecular cations. E.g: n-pentane, the ion at m/z=72 is smaller than fragment cation at m/z=43. This indicates that stable fragments dominate the spectrum, providing information about the molecule's most stable parts.

M+1 peak	M+2 peak
The M+1 peak appears in mass spectrum because of presence of ^{13}C isotope in molecule.	M+2 appears in mass spectrum due to presence of heavier isotopes like ^{37}Cl in molecule.
The M+1 peak is common in all organic compounds contains Carbon atoms.	The M+2 peak appears generally in compounds containing elements with

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	significant heavier isotopes like Cl or bromine.
M+1 peak helps to calculate the no of carbon atom in molecule.	M +2 peak helps to identify the presence of Chlorine or bromine atoms in molecule because of characteristics peak ratios.

Chp 18: Spectroscopy

c) Calculate the degree of unsaturation in

i) Benzene (C6H6):-

Molecular Formula of benzene = C6H6}

No of Carbon atoms = 6, no of hydrogen atoms = 6, no of halogen atoms = 0

Degree of unsaturation = (2C+2-H-X)/2

Degree of Unsaturation = (2 × 6 + 2 - 6 - 0) / 2 = 4

So the degree of unsaturation for benzene is 4 (three double bonds and one cyclic ring.

ii) Ethyne (C2H2):-

Molecular formula = C2H2

No of carbon atoms = 2, no of hydrogen atoms = 2, no of halogen atoms = 0

Degree of unsaturation = (2 × 2 + 2 - 2 - 0) / 2 = 2

So, degree of unsaturation for ethyne is 2.

(2) a) Define IR Spectroscopy? b) What is the relation of dipole moment with IR radiations absorption in IR spectroscopy?

Ans:- a) IR spectroscopy uses the interaction of infrared radiations with organic molecules. It is typical range of wave number from 4000 $\times cm^{-1}$ to $625cm^{-1}$. It is used to analyze functional group present in organic compounds.

b) When an organic molecule with a functional group that has a hetero atom is exposed to IR radiations, the electric field part of rays interacts with dipole moment of molecules. When vibrational frequency of a bond or group of atoms inside a molecule matches the natural frequency of infrared light, absorption take place. This start bonds to vibrate with greater amplitude after absorbing energy.

(3) a) Define UV-Visible spectroscopy? b) Relate absorption of visible light with colour of compound?

Ans:- a) UV-visible spectroscopy is a technique in which absorption of UV and visible light by compound is measured. Its typical wavelength range is from 200\text{ nm} to 800\text{ nm}. Absorption of light is directly proportional to concentration of sample available when it has electrons for excitation.

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b) Coloured compounds absorb only a small range of wavelength (400\text{ nm} to 800\text{ nm}) in visible region of electromagnetic radiations and transmit others. which makes them coloured. The colour of compound is not predicted by wavelength it absorbs, but by the wavelength it reflects or transmits.

(4) a) A compound absorbs light with wavelength of 500- 560nm. What colour do you expect for this compound?

Ans. From table, the range absorbed corresponds to 500\text{-}560\text{ nm} is green colour. Thus colour absorbed is Green. Since complementary colour is actually observed. Thus, the compound will appear purple.

(5) Differentiate b/w Atomic Emission spectroscopy and Atomic Absorption Spectrum?

Atomic Emission Spectroscopy	Atomic Absorption spectroscopy
In AES, atoms first excited to higher energy level by heat.	In AAS atoms remain in ground state and absorb wavelength of light to higher energy
The instrument measures light emitted by atoms which appears as bright lines on dark background	The instrument measures the light absorbed by atoms which appears as dark lines on bright background.
AES is mainly used to detect the presence and concentration of element by their emission intensity..	AAS is mainly used to measure the concentration of specific elements by amount of light absorbed.

(6) a) What is NMR? b) Write the principle of NMR and what type of nuclei absorbs radiations in NMR spectroscopy?

Ans:- a) It is spectroscopic technique based upon the re-orientation of atomic nuclei that have non-zero nuclear spins to an external magnetic field. This happens when electromagnetic radiations is absorbed in radio frequency region, from about 4 to 900MHz. The Frequency depends on isotopic type of nucleus and increases with strength of magnetic field.

b) All NMR active nuclei have spins same with random orientation.

* When an external magnetic field applied, these nuclei acquire two spin states: i) At lower energy state, aligned with applied magnetic field.

ii) At high-energy state, opposite to applied magnetic field.

All NMR active nuclei have a property called Spin. A charged nucleus spins about an Axis and creates its own magnetic dipole moment. e.g:- Hydrogen, ^{13}C isotope of carbon, ^{19}F isotope of fluorine have magnetic moments and can be studied by NMR.

(7) a) How many peaks do you expect in proton NMR of benzene? b) Explain the chemical shift?

Ans:- a) Benzene (C6H6) is highly symmetrical molecule. All six hydrogen atoms are chemically equivalent. In proton NMR, equivalent protons give same signal. Therefore, all six protons produce single signal. Thus only 1 peak around 7.26pp} will be observed.

b) Chemical shift is the difference b/w resonant frequency of nucleus and standard in magnetic field. It is measured in ppm. It shows the electronic environment of nucleus. Downfield peaks show deshielded proton and upfield peaks show shielded ones.

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(8) What is deuterium exchange method and what is its use?

Ans:- When sample containing -OH or N-H protons is shaken with D2O, these protons exchange with deuterium which causes disappearance of their peaks. This method is used to confirm presence of O-H or N-H group. Deuterium is NMR inactive so it does not give signal in NMR.

(9) What is TMS? Tell its significance?

Ans: TMS is Tetramethylsilane (Si(CH3)4). All its protons are chemically equivalent and strongly shielding by electrons. TMS gives a single sharp peak which is used as reference at δ= 0 ppm. It is used to calibrate chemical shift scale in NMR spectroscopy.

(10) a) Why deuterated solvents are used in NMR spectroscopy? b) What is the difference b/w ^{1}H -NMR and ^{13}C-NMR?

Ans:- a) Deuterated solvents such as D2O or (CD3)2CO are used so that solvent protons do not interfere with sample peaks. Deuterium is NMR inactive, so no signal is produced by solvent. This allows clear observation of proton peaks of compound being studied.

^{1}H -NMR	**^{13}C-NMR**
1) It gives information about hydrogen atoms in a molecule.	1) It gives information about Carbon atoms present in molecule.
2) It shows no of different types of molecule. Such as hydrogen	2) It shows no of different types of carbon atoms in molecule.
3) ^{1}H-NMR} is more sensitive and gives stronger signals.	3) ^{13}C -NMR} is less sensitive because it is only about 1.1% Abundant naturally.

Chp:19 Chromatography

(i) What cause color to separate during chromatography?

Ans:- Color separate because different components of a mixture have different affinities for stationary phase and the mobile phase. Some components stick strongly to stationary phase and move slowly while others move faster with mobile phase. The difference in movement separate the mixture into distinct spots or bands on chromatogram and provides valuable information about composition and purity of compounds.

(ii) Explain possible use of thin layer chromatography?

Ans:- It is use to separate substances based on their interactions with stationary phase. The stationary phase is a thin layer of adsorbent such as silica gel or alumina coated On a plate mixture substances. Strongly that stick more to Stationary phase while other move faster with mobile phase This separates the mixture into different components that appears as spots. These spots can be compare al with known standard for identification. It is inexpensive, widely Used in Forensic for pharmaceutical, food, and industries-

(3) Differentiate b/w TLC and Paper chromatography?

Thin Layer Chromatography	**Paper Chromatography**

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TLC provides better resolution and cleaver separation of spots.	P.C provides lower resolution and spots may be broader
TLC is widely used in industries. Chemical analysis, and forensic applications. due to its efficieny.	P.C used for simple preliminary analysis and teaching purposes.
TLC can separate very small quantities of sample quickly and efficiently.	It requires relatively larger sample quantities and is less Sensitive.

(4) a) What type of Chromatography is Used in Forensic sciences? b) Assess the limintations of paper chromatography when used to separate complex mixture?

Ans:- a) TLC separates and identifies drugs like cocaine or heroin by comparing spots with known standards. Used for poison detection in blood or crine samples, trace evidences analysis such as fibers and points, gunshot residue detection, explosives detection, and examination of ink and paper in documents.

b) It is suitable only for detecting very small quantities of substances. Spots can spread and overlapp making identification difficult: Paper chromatography are slower and less precise than TLC.

(5) What are essential characteristics of substances used as developer?

Ans:- A developer is used in chromatography to make colourless spots visible after separation.

1- It should react specifically with compounds of interest to Produce a visible colo..

2- It must produce a clear and sharp spot without Smearing or spreading.

3- It should not damage the Stationary phase or separated components.

4- It should be easy to handle, safe and redily available-

(6) what factors affect the retention time in column chromatography How do polarity and boiling point influence it?

Ans:- Retention time is time a molecule takes to pass from injection to detection

Non-polar components have longer retention times because they interact strongly with non-polar stationary phase. Polar compounds interact less and move faster giving shorter retention times.

* Molecules with longer boiling points vaporize more easily reducing retention time. difference in retention time allow separation and identification of mixture component.

(7) Is it possible for Rf value to exceed 1/

Ans:- No, Rf valve cannot exceed 1. It is Calculated as:-

$$Rf=\frac{Distance\ from\ baseline\ travel\ by\ solute\ compound}{distance\ from\ base\ line\ travel\ by\ solvent}$$

* Since the solute cannot move farther than solvent front. Rf is always less than 1.

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Chp: 20 Material

(1) Compare and contrast the properties of plastic & ceramics? Comment on environmental impacts of using plastics and steel in construction?

Ans a) Plastics have lower density, are Chemically inert good insulators and can be moulded into many shapes They are light in weight and low cost.

* Ceramics are very hard and brittle having stitness and strength compare to metals. They are poor conductors of heat and electricity and may be transparent translucent Or opaque. Plastics are flexible cheap while ceramics are riged and good for thermal insulation.

b) Steel is Strong, durable, recyclable so it reduces waste by being reused many times. Plastics are lightweight and chemically stable which helps save material and energy. But plastics are non-biodegradeable and Cause pollution if not recycled properly, Using both materials requires safe recycling practices to avoid toxic effects.

(3) a) Describe the properties and Uses of Ceramics? b) What is the process of smelting in metal extraction?

Ans:- Ceramics are very hard, brittle and have stiffness compare to metals. They are poor conductors of heat and electricity, so they are excellent thermal insulators. Ceramics can be transparent translucent Or opaque. Common ceramics are used to make tiles, cement glass pro porcelain and even iron- Oxides material like Strontium magnetic magnets.

b) Smelting is method of reducing ore with reducing agents by heating to free metal ions Common reducing agent are cake, Carbon monoxide and Hydrogen.

$$ZnO + C(s) \rightarrow Zn(s) + CO(g)$$

This process produces molten metal separated from impurities.

(4) a) Explain the role of catalyst in chemical reaction? b) What is Ore concentration in context of processing?

Ans: a) A catalyst speeds up chemical reaction by lowering Activation energy. It provides an alternate pathway for the rnx but is not consumed and can be reused. Catalysts may form temporary Unstable substances during reaction.

$$2H2O2 \rightarrow 2H2O + O2$$

* These Reactions occurs faster at Same temperature.

b) Concentration means removing impurities from the ore to increase metal content. Common techniques include flotation process, magnetic separation, and gravitational separation. This step makes ore ready for roasting Or smelting.

(5) Lists four applications of X-ray crystallography?

Ans:- 1) Used to determine structure of RNA and DNA

2) Study the atomic structure of materials like polymers. 3) Used to reveal target protein binding sites for new drugs and to study atomic structure of materials.

(6) How does refining improve quality of metals?

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Ans: Refining removes impurities from metals obtained after smelting.

* In electrolytic refining, impure metal act as anode and pure metal act as cathode. e.g: In copper refining the Cu^{2+} ions deposit on the cathode and pure copper while impurities like Au and Ag fall as anode mud This makes metal 99% pure or more.

(7) what is difference b/w smelting and Refining of metals?

Ans: Smelting reduces metals ion in ore to Free metal using heat and a reducing agent.

e.g:- $Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$

* Refining removes impurities to produce pure metal. Electrolytic refining uses an impure metal anode and pure me tal cathode Impurities like gold and silver Falls as anode mud.

Both steps are needed for pure, usable metals.

(8) What are main challenges in modern material recycling

Ans:- Recycling is challenging because modern materials have multiple layers or may be complex alloys. Contaminants like oils and adhesives reduce purity. Some materials contains toxic substances like lead mercury. Recycling be costly and may require advanced technology. sofe recycling needs proper sorting, testing and handling.

(9) How does Bessemerization process help in producing pure metals?

Ans: In Bessemerization, air is blown through molten copper matte. Ion sulfrae Oxidize to FeO

$$2FeS + O_2 \rightarrow 2FeO + 2SO_2$$

FeO react with sand to form $FeSiO_3$ slag. Remaining cuprous sulfide reacts with curprous oxide to form metallic copper.

$$Cu_2S + 2Cu_2O \rightarrow 6Cu + SO_2$$

* This produces blister copper, 97-99% pure

Chp: 22 Agriculture

(1) What are Insecticides? How do they work?

Ans:- **Insecticides** kill or control insects. Organophosphates and carbamates block acetylchloinesterase causing nervous Overstimulation and paralysis of insects.

Pyrethroids disrupt sodium channels, using repetitive nerve impulses and paralysis.

* **Herbicides**: Herbicides control unwanted plants (weeds) - Glyphosphate inhibits protein production and kills broadleaf weeds and grasses.

2,4-D and Dicamba Overstumilate the plant hormone auxin, causing uncontrolled growth and death.

*** Fungicides:** Kills fungi or prevent their growth Triazoles

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(2) a) What are the benefits of pesticides? b) What risks are associated with Use of pesticides?

Ans : a) It protects plants from pests and diseases increasing yield.

* Pesticides reduce disease Spread among plants animals and humans.

* Pesticides improve fruit and vegetable quality.

* b) Pesticides can be neurotoxic, affecting famers, workers and consumers.

* Pesticide residues Can affect consumers.

* Pesticides reduce beneficial insects and biodiversity.

(3) Describe the chemical reactions involved in soil acidification due to acid rain?

Ans:- Acid rain lowers the soil pH by reacting with soil minerals.

Example: $CaCO3 + H2SO4 \rightarrow CaSO4 + H2O + CO2$

* Removes calcium from soil.

* It also cause aluminium leaching which is toxic to plants.

$$Al2O3 + 3H2SO4 + 6H2O \rightarrow 2Al(OH)3 + 3SO4^{2-} + 6H^{+}$$

(04) Explain the role of phosphorus in plants growth and development?

Ans: Phosphorus helps root growth flower and seed formation. It is part of ATP for energy transfer. It also improves plant maturity and yield.

(05) a) How does Acid rain affect nutrient uptake by plants? b) Define Acid rain? What are the impacts of Acid rain?

Ans: a) Acid Rain lowers soil pH and removes Ca^{2+}, Mg^{2+}, K^{+}, Na^{+}. These nutrients are replaced by H^{+} ions. Plants get fewer nutrients, so growth is poor.

b) Normal rain has pH 5.6 due to dissolved CO_{2}. Rain with pH less than 5.6 is called Acid Rain.

IMPACTS OF ACID RAIN:

* Acid rain lowers pH of soil, making it more acidic.

* Acidic soil loses nutrients such as Ca^{2+}, Mg^{2+}, Na^{+} replaced by H ions.

* Acid rain can damage leaf surfaces increasing water loss and disease susceptibility.

(06) a) Describe the chemical composition of super phosphate fertilizer and its function in soil fertility? b) what is meant by "pesticide resistance" and why is it a concern?

Ans: a) Superphosphate is made by treating phosphate rock with sulfuric acid to form $Ca(H2PO4)2 + CaSO4$. It provides soluble phosphorous to plants. It improves root growth and crop productivity.

b) Pesticides Resistance is when pests survive after pesticide use. They become harder to kill. This cause farmers to use more chemicals, harming environment.

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(07) a) Explain the impact of acid rain on soil microbial activity? b) Explain the process of genetic engineering in agriculture.

Ans: a) Acid rain affects soil microbes and reduces nutrient cycling. This changes soil structure and fertility. Less microbial activity lowers crop growth.

b) Genetic Engineering changes organisms genes to improve traits.

Steps:

* Identify trait

* Isolate gene

* Insert into plant genome

* Grow and test plant

This makes crops better in yield, nutrition and resistance.

(8) a) Discuss one human health concern associated with pesticide exposure? b) Explain the difference b/w organic and inorganic fertilizers and their impacts on soil?

Ans: a) Pesticides can cause nervous system damage, skin problems or cancer. Long-term exposure is dangerous for farm workers and nearby residents.

b) Organic fertilizers: Release nutrients gradually as they decompose, improving soil health in long term. Inorganic Fertilizers provides nutrients rapidly for quick plant uptake. While both can help plants, inorganic fertilizers can reduce soil carbon, harm beneficial organisms and cause acidification. Organic fertilizers maintains ecosystem balance.

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